

REVIEW

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Large language models for clinical artificial intelligence in healthcare a systematic review

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Abstract

Large Language Models (LLMs) have demonstrated the capacity to process, reason, and generate extensive volumes of data, providing a novel paradigm for integrating generative artificial intelligence (GenAI) into the medical field. Multimodal LLMs (MLLMs) extend these capabilities by incorporating diverse data modalities into unified representations, including genomics, medical imaging, and clinical text. This systematic review synthesizes advancements from 246 records identified between January 2020 and September 2025, of which 90 studies were included after full-text screening, to address critical gaps in understanding the clinical role of LLM and MLLM in healthcare. We trace the evolution from classical natural language processing (NLP) approaches to modern transformer-based architectures, summarize their technical foundations, and examine their construction, evaluation, and deployment in medical workflows. Key contributions include highlighting multimodal integration (e.g., imaging-genomics-text fusion), ethical governance frameworks, and validated domain-specific fine-tuning in clinical settings. We also highlight advances in Prompting, Retrieval-Augmented Generation (RAG), and Multi-Agent (agentic) workflows, providing a critical assessment of their benefits and limitations. In addition, we analyze challenges such as hallucinations, bias, and privacy risks, while providing actionable guidelines for clinicians, developers, and policymakers to improve regulatory compliance. By consolidating the nomenclature and systematically evaluating GenAI in medicine, this review offers evidence-based recommendations and directions for the safe and effective integration of generative AI into healthcare. The findings are intended as an authoritative guide for researchers and practitioners, bridge principles, clinical applications, and policy considerations for LLM and MLLM.

Keywords Large language models (LLMs), Multimodal large language models (MLLMs), Generative artificial intelligence (Gen AI), Model hallucinations, Regulatory compliance, Natural language processing (NLP), Generative AI in medicine and healthcare

1 Introduction

With rapid integration into clinical practice, Large Language Models (LLMs) are becoming a presenter of the transformation through their capabilities across many medical applications. Foundation models like OpenAI's GPT, Google Gemini, and LLAMA can



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comprehend and produce text corresponding to human language to improve patient care, optimize clinical workflows, and assist healthcare professionals in decision-making. Patient monitoring, diagnostic assistance, clinical decision making, and therapeutic interventions are the main applications of LLM in clinical practice. In the following, we examine these applications in healthcare.

Patient Monitoring and Interaction LLMs can monitor chemotherapy-induced toxicities by classifying subjective patient-reported symptoms, achieving precision comparable to expert oncologists. This application can improve patient monitoring and reduce the workload of healthcare professionals by enabling early interventions and improving quality of care [1]. In aging and dementia care, LLMs offer the potential for precision medicine and individualized treatment strategies, providing clinical decision support and predictive analytics to improve patient outcomes [2].

Diagnostic and Clinical Decision Support LLMs are being explored as clinical decision support tools, particularly in oncology, where they help identify guideline-based care for conditions such as colon cancer. However, current models often provide verbose responses with missing or inaccurate information, indicating a need for further optimization [3]. In mental health, LLM can help detect, diagnose, and treat stress and trauma-related conditions, providing just-in-time interventions and alleviating the burden on human professionals [4]. In clinical documentation and note generation, LLMs can automate the generation of clinical notes, reducing the administrative burden on healthcare professionals and allowing them to focus more on patient care [5]. These models can also help compile patient notes and help patients navigate the healthcare system. However, their use is limited by biases in training data and the generation of inaccurate information [5]. LLMs can accurately detect diseases from EHRs, outperforming traditional methods in identifying conditions such as gout and calcium pyrophosphate deposition disease (CPPD) with high precision and predictive values [6]. In clinical decision support, LLMs help in clinical decision making by providing expert level competence in medical questions and facilitating the understanding of complex medical information [7]. On the other hand, diagnostic assessments are used in particular in mental health, where language plays a crucial role in understanding and treating conditions [4].

Therapeutic Support and Patient Engagement LLMs have been applied to therapeutic support, responding to patient questions, and generating clinical summaries. They can also improve patient participation by offering personalized support and improving chronic disease management [8]. In gastroenterology, LLMs have the potential to transform clinical workflows and improve patient outcomes, although critical barriers to implementation remain [9].

Integrating large-language models within clinical practice faces many challenges, mainly in producing inaccurate information, inherent biases, and ethical issues, including privacy and data security. Due to misinformation attacks, slight alterations in LLMs can lead to the dispensation of wrong biomedical information, which makes them increasingly potentially massive in trustworthiness apprehension within healthcare settings [10]. Moreover, LLMs are usually prone to cognitive biases, which have an effect on their ability to be accurate in clinical situations; therefore, more robustness and effective bias mitigation strategies are one of the critical ways for reliable applications in healthcare [11]. Ethical concerns are further compounded because LLMs handle sensitive medical information, so patient data must be safeguarded with privacy-preserving

techniques [12]. Another major hindrance to their broad adoption in clinical practice is the lack of standardized evaluation of LLMs. Current human evaluation practices for these models are not sufficiently reliable and generalizable to have required development that has been developed in frameworks such as QUEST to ensure LLM safety and effectiveness within health care [13]. The ethical considerations of LLM include, but are not limited to, transparency, fairness, and health equity; each requires ongoing review and governance on a case-by-case basis [14]. Fine-tuning large language models and retrieval-augmented generation can improve precision and applicability in a clinical setting for specific challenges [15]. Some studies have explored how to adapt large language models for clinical Information Extraction (IE). Vithanage et al. [16] evaluated LLaMA 3.1-8B on named entity recognition tasks in residential aged care facilities, comparing zero-shot and few-shot learning with PEFT and RAG. They found that PEFT significantly improved performance in both learning settings, while RAG offered benefits primarily in few-shot learning. These findings offer practical guidance on tailoring LLMs to healthcare use cases with limited labeled data. However, a responsible introduction into practice calls for multidisciplinary participation of stakeholder in concerted efforts that will include detailed procedures for safety, oversight, and transparency while promoting innovation in that field [17]. In general, LLMs have much promise in improving clinical decision-making, but addressing these complex challenges is vital to ensure their effective and ethical deployment in healthcare.

This survey aims to thoroughly examine recent advancements in medical LLM from diverse perspectives. We carefully collect the relevant literature and synthesize the foundational context, which includes core techniques, knowledge, algorithms, implementation strategies, and the challenges associated with Medical LLMs.

As illustrated in Fig. 1, this paper aims to address the advancements that Gen AI, especially LLMs and MLLMs, have achieved, elaborating on development trends, applicable domains together with integration into the practice of modern medicine, LLMs having been preceded by paradigm shifts in the sociology of LLM led by early NLP models and the deep learning era that was later followed by a surge of more sophisticated LLMs. In this paper, we will focus on the structure of these models, including single types such as encoder, a single type of decoder, and two types of encoder-decoder structures. We will also include multimodal models that allow the correlation of different data types. With medical diagnosis, clinical report generation, medical, educational, and mental health services, among others, it is apparent that LLM and MLLMs can transform clinical workflows. The evaluation aspect, by definition, integrates the performance evaluation metrics of automatic evaluations, human evaluation, and task performance evaluation benchmark evaluation modes, among other assessment models. However, there are issues of hallucinations, privacy and security issues, bias and fairness issues, and the issue of high cost of training that are deep rooted and require a lot of effort to be wider in scope. The paper highlights future directions, emphasizing the importance of multimodal integration, trust in Gen AI in the medical field, and development to address these challenges and advance the integration of LLMs and MLLMs into the medical domain. This paper examines the relevant issues related to the growth and deployment of LLMs and MLLMs in the healthcare domain with examples from practice. What are the main structural developments in their growth and what new concepts of pretraining, fine-tuning, and integrating modality have made them better and more flexible? How are

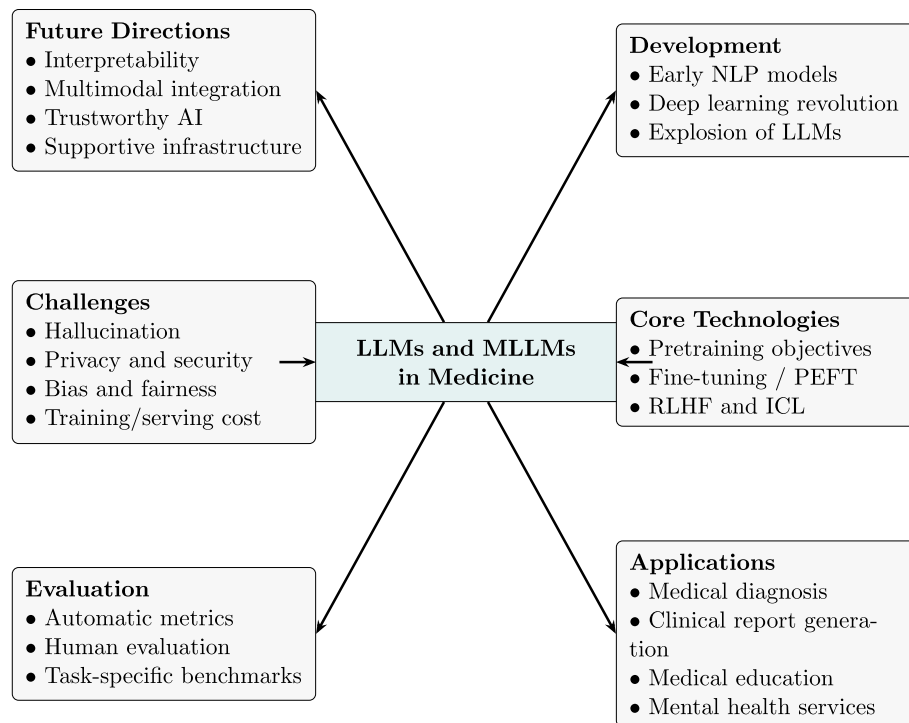


Fig. 1 Proposed review architecture for LLMs and MLLMs in medicine

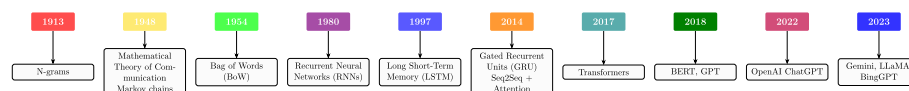


Fig. 2 Evolution of language models timeline

these models used today in the field of medicine—with a focus on the models about medical diagnosis, clinical report writing, and educational and instructional purposes within the medical sphere—what these still do not cover, for example hallucination, protection of personal data, bias, etc.? Finally, what measures and infrastructure are required to enhance interpretability, establish reliable AI frameworks, and aid the controlled and efficient integration of such systems in the healthcare domain? The authors seek solutions to these questions to provide functional recommendations to the research, clinical, and policy communities.

2 Background and related work

This section presents a summary of language models showing the advances throughout the historical trajectory of language models. This evolution began with the advent of fundamental language models, including n-gram and rule-based models [18]. Subsequently, the landscape of language models became mainly characterized by neural language models, which were primarily inspired by techniques in neural machine translation [19]. Over time, increasingly effective and essential components, such as the attention mechanism [19], word embedding [20], and positional encoding [21], were integrated, ultimately leading to the emergence of sophisticated architectures like the transformer [21]. The timeline depicted in Fig. 2 illustrates the development of language models. A timeline of the evolution table of NLP to LLM (1) illustrates how modern architectures

like GPT-4 and Med-PaLM 2 transcend classical rule-based systems, allowing applications from automated diagnosis to patient participation. (Table 1)

2.1 Natural language processing foundations

Language models were originally started early by Markov models [18] and n-grams [22] as the mathematical foundations of Markov models served as the basis for n-grams, which are used to create chains to explore the probabilities of the next vowels and consonants. The concept of the first n-gram was introduced in the works of Markov [18]. Later, Claude Shannon used Markov processes to propose a model for natural language [22], where the nth order of Markov chains is used to build a statistical model that would represent the probability of sequences of letters, such as words and sentences. The idea was to calculate the frequency of length n's character sequences, n-grams. In [22], he laid the foundation for language models and, more importantly, the foundations of telecommunication. The fundamental works of Markov. Later, several approaches in modeling language have come into existence, including rule-based systems, neural network-based systems, and recently trained large language models (LLMs) built according to the architecture of the Transformer [21] and the attention mechanism [19].

2.1.1 Vector space and word embeddings

Vector space models represent linguistic items (words) as vectors in particular vector spaces. Word embedding refers to the expression of words in vectors (also called vectorization of words) [23]. In addition, bag-of-words (BOW) is one of the simplest word embedding approaches that map words to associated vectors. The idea of word representations in vector spaces was mainly brought about by several other developments, first by [24] and then in Word2Vec [25]. These techniques became integral to language modeling and were widely adopted in various applications. For example, [26] applied temporal word embeddings to detect psychological disorders from social media data, demonstrating how changes in word usage over time could be used for early detection of mental health risks, even without large generative models.

2.2 Deep learning revolution

Artificial neural networks (ANN) have been used for natural language processing and the development of other models of machine learning and artificial intelligence. We are reviewing some of the most important and foundational ANNs, such as RNNs and LSTMs, for more advanced and modern models, such as attention-based models and transformers. The deep learning revolution was crafted with the powerful backpropagation algorithm for neural networks and embeddings inspired by Word2Vec, among

Table 1 Evolution of AI models in medical applications

Era	Model type	Primary medical applications	Key limitations
Pre-2010	Rule-based NLP	ICD-10 coding, keyword extraction	Sensitivity to linguistic variability
2010-2017	RNNs, LSTMs	Clinical note parsing	Short-term context retention
2018-2021	BERT, GPT-2	EHR summarization, sentiment analysis	Single-modality constraint
2022-2024	GPT-4, Med-PaLM 2	Multimodal diagnosis, digital patient twins	Hallucinations, regulatory ambiguity
Total eras	4 milestones	Expanded clinical use cases	Persistent challenges

The table outlines key eras in the progression of AI models used in healthcare, emphasizing their technical evolution, clinical roles, and known limitations

others. Such advances led to the development of Transformer architectures, the new paradigm shift in NLP capabilities. Through their attention mechanisms, transformers empowered models with the ability to process huge datasets, hence creating large language models (LLMs) that understand intricate medical terminologies. Convolutional and transformer-based models have been widely adopted for medical imaging tasks. For example, in the detection of retinal disease, OCT images were classified using a compact convolutional transformer (OCCT) model, with generative adversarial networks (GAN) used to augment underrepresented classes [27]. Deep learning has also been applied to biosignal data for the prediction of neurological events. [28] proposed both supervised and unsupervised deep learning models to distinguish preictal (pre-seizure) EEG from normal EEG in a person-specific manner. Oh et al. [29] demonstrate how cluster-based deep reinforcement learning can personalize treatment recommendations, aligning with data-driven change in clinical AI.

2.2.1 Recurrent neural networks (RNN)

were first developed in the early 1980 s [30] as a type of artificial neural network designed to process sequential and time series data with arbitrary lengths rather than fixed input. Recurrent Neural Networks (RNNs) process sequential data (e.g., patient histories) by iteratively updating hidden states. However, their inability to maintain long-term dependencies, such as linking prior chemotherapy from a patient to current symptoms, led to innovations such as LSTM gates. For example, BioLSTM [31] improved mortality prediction in ICU patients by capturing temporal patterns in laboratory results. Word embeddings such as Word2Vec later enabled semantic understanding of medical jargon (e.g. mapping myocardial infarction and heart attack to proximate vectors), but struggled with rare terms such as Charcot-Marie-Tooth disease [32]. Although RNNs are useful for improving language models, they still need to perform better for processing series with long-term dependencies [33]. One limitation is the vanishing gradient [34] and the explosive gradient [35], which have encouraged researchers to design more suitable solutions that can handle long sequences.

2.2.2 The long short-term memory (LSTM)

Hochreiter S [36] was proposed in 1997 to detect longer-term patterns in the input sequence and the vanishing gradient problem in RNNs. The LSTM network has a memory cell and input, output, and forget gates. The forget gate provides long-term or short-term memory to control the propagation of information in the network. Although LSTM networks can handle the RNN's shortcomings, LSTM networks are complex. Hence, computationally more expensive. The GRU [37] proposed a simpler architecture with fewer parameters than the LSTM to accelerate training convergence by reducing gates and solving the problem of the disappearance gradient in RNN. However, LSTMs and GRUs have strengths and weaknesses, making them suitable for different settings.

2.2.3 Sequence to sequence (seq2seq) language models

One of the main driving forces for the development of this class of language models has been the field of machine translation [38]. Mostly, machine translation tasks involve general sequence-to-sequence models [39] or solve problems involving sequence modeling. In these applications, the goal is to develop models that are able to relate a sequence in

one domain to a suitable sequence in another domain, e.g., translation of a sentence from one language to another language. The design of the Seq2Seq model is fundamentally based on encoder-decoder architectures [39], in which both the encoder and decoder are derived from RNN-based architectures such as LSTMs [36] and GRUs [37], respectively. In this modeling approach, the encoder reads the input sequence first and creates a vector representation of the input, namely, the context vector. The decoder then takes the provided context vector and generates the output sequence. Models of this nature are widely utilized for tasks related to machine translation, operating on RNN-based frameworks such as LSTMs or GRUs. This dependency inherently leads to the adoption of their limitations, including challenges in managing long-range dependencies [40]. Furthermore, a significant avenue for improving Seq2Seq architectures lies in their treatment of sequences, as they tend to process these sequences without considering the varying importance of different components within them [38]. As a result, Seq2Seq models must be enhanced in their attention to important elements of a sequence, such as keywords and crucial segments of a sentence that convey important information. This limitation prompted researchers to develop methods through which models could adaptively concentrate on certain areas of a sequence [38], which ultimately led to attention-based language models and, later, the transformer model [41].

NLP has become a computational approach to understanding and generating human languages. The early models of RNNs and LSTMs lay at the core of machine-based language comprehension. In the medical field, the early model's ability to scale could have been higher, as well as the contextual understanding. Hence, this seriously limits their application in vast domains such as medicine. Hence, recent generative LLMs open the door to a greater impact in the medical field.

2.3 Language models revolution

2.3.1 Attention-based language models

Adding an attention mechanism to encoder-decoder architectures was a real game changer to satisfy the missed part of the puzzle in the previous latest enhancements and research mentioned above. The idea was that the relative importance of the elements of a sequence can also be encoded and assigned to each element while the sequence is being processed. The attention technique proposed in [19] uses an alignment model that computes a score. Generate some weights by passing the calculated alignment score to a softmax function. Then it is used to get a weighted sum of the hidden states of the encoder to obtain the wanted context vector. The attention mechanism has been important for the development of artificial neural networks and machine learning models. More concretely, it triggered various other influential findings on natural language processing, including the famous transformer architecture [21]. The transformer architecture followed with the rise of innovations such as the BERT model [42], GPT models [43]. With the rise of such models, NLP underwent rapid evolution. Most of these models, trained on diverse datasets, achieve unparalleled accuracy in text comprehension, reasoning, and generation. That has made them even more scalable, thus becoming prominent players in integrating artificial intelligence (AI) with different fields including medicine. Attention mechanisms act like the focus of a radiologist on the edges of a tumor in an X-ray, ignoring irrelevant regions also using multimodal models like in [44].

2.3.2 The transformer

The transformer architecture was introduced in 2017 by Google Research [21]. The transformation architecture is an encoder-decoder architecture [45] that does not use recurrent layers with only attention mechanisms and word embeddings (embedding layers), dense layers, layers, and normalization layers. Because it is not recurrent, this helps to avoid vanishing or exploding gradients (the RNN's drawbacks), and it can also capture long-range patterns than RNNs. It enables data to be processed in parallel across multiple GPUs, allowing faster training. Transformers can be used in revolutionary NLP techniques like multimodal large language models with text and images. Transformers process data in parallel, as if a medical team is simultaneously analyzing MRI, genomics, and patient notes. MMLLMs will be discussed in the upcoming sections.

2.3.3 Large language models (LLMs) avalanche

With larger transformer-based architectures trained on huge datasets, the progress of creating these large models accelerated. The remarkable advancement of large language models (LLMs) within the Natural Language Processing (NLP) society, characterized by a series of seminal advancements, such as BERT [42] and Gemini [46] passed through different evolutions from early NLP foundations to deep learning revolution reaching to large language model explosion. GPT paper [43] by Open AI researchers and BERT [42] by Google demonstrated the effectiveness of unsupervised pretraining using the transformer-based architecture. Some transformer modules based on the basic transformer's decoder trained on a huge dataset to predict the next token using self-supervised learning and then finetuned it on several language tasks, including text classification, similarity, and question answering by answering some multiple-choice questions. This led to a new model in NLP, that is, unsupervised pre-training and fine-tuning [47, 48].

Taking advantage of the scalability of the Transformer architecture, train large unsupervised language models on large datasets without labels through techniques like masked language modeling (MLM) or next-single prediction (NSP) [42], and then fine-tune for specific applications with relevant supervised objectives [43]. We can get the following advantages of this framework: (1) the pre-training dataset can be created from any set of unlabeled text, in this way removing the requirement that supervised learning has for large annotated datasets [49]; (2) training based on large amounts of unlabeled data helps to develop better, more general, and abstract language representations, which, as a result, provides greater generalization; (3) during the fine-tuning stage, during which one adapts to downstream tasks, it will be sufficient to formulate just the relevant objective functions.

To better understand the landscape of large language models, their applications, data requirements, evaluation strategies, LLM training methods, key players in LLM, and more, a comprehensive summary is illustrated in Fig. 3. This conceptual map captures the interrelationships between key aspects of LLM development and deployment, providing an integrative view of the current state of research and practice.

2.3.4 From text-only to multimodal

Although large language models (LLMs) have revolutionized text-based medical tasks such as symptom analysis and clinical decision support, deep learning techniques are advancing the field of medical image retrieval [50, 51]. ProgressLLM research led to the

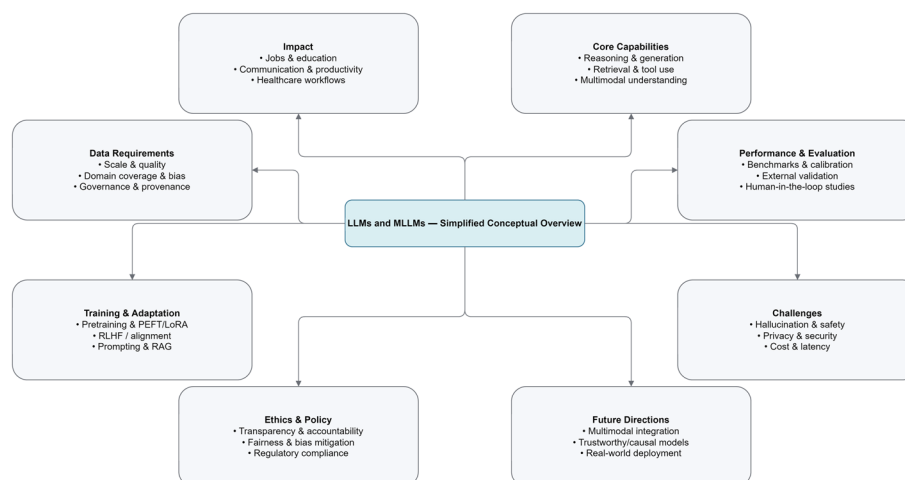


Fig. 3 Simplified conceptual overview of LLMs and MLLMs. The figure summarizes key themes (impact, core capabilities, data requirements, performance, training, challenges, ethics, and future directions)

emergence of a series of outstanding works such as PaLM [52] and LLaMA [53]. However, these LLMs are still only capable of understanding a text. Flamingo [54] can adjust to visual assignments using few-shot learning methodologies without necessitating task-specific fine-tuning. Subsequently, more prominent works emerged in the multimodal domain, such as LLaVA [55], and GPT-4o [56], all of these methodologies use the fundamental objective of incorporating a visual encoder into large language models (LLMs) while employing supplementary modules to facilitate the integration of LLMs and visual encoders, thereby addressing the disparity between distinct modalities. Recent works also explore vision-language models tailored for medical image segmentation, such as RSKD by Liang et al. [57], which incorporates rank-sensitive knowledge distillation.

2.3.5 From general-purpose LLM integration to clinical applications

Recent surveys, including those by Yang et al. [58] and Annepaka and Pakray [59], have provided comprehensive overviews of the development, capabilities, and applications of Large Language Models (LLMs). Yang et al. [58] focus on the integration of LLMs with structured knowledge sources such as Knowledge Graphs and Retrieval-Augmented Generation (RAG), offering a broad perspective on knowledge-enhanced LLM architectures across general domains. In parallel, Annepaka and Pakray [59] trace the historical evolution of LLM, detailing key architectures, training paradigms, and their impact across diverse sectors, including business, education, agriculture, and healthcare. They emphasize the challenges of keeping up with the rapid advancements in the field, underscoring the need for comprehensive summaries that span foundational principles, societal impacts, and practical applications.

Building on these foundational works, our research narrows the focus to the healthcare and clinical domain, where the deployment of LLM and Multimodal LLMs (MLLM) introduces unique challenges. These include concerns related to data privacy, medical bias, and regulatory compliance, as well as the complex integration of multimodal data sources such as genomics, imaging, and clinical notes. By synthesizing insights from more than 1000 recent medical studies, we trace the trajectory of LLM adoption in healthcare, explore domain-specific fine-tuning techniques, and highlight practical hurdles in real-world deployment. In doing so, our review complements the broader LLM

literature with a domain-specific perspective intended to guide clinicians, researchers, and policymakers working in medical AI.

3 Medical large language models (Med-LLMs)

Med-LLMs have the potential to be a game changer in the health field by transforming the delivery of health care and its outcomes. Various models such as ChiMed-GPT, MedicalGPT, HuatuoGPT-II, and ChatMed seek to simplify medical knowledge, improve diagnostic abilities, and offer customized treatment options to patients [60, 61]. For example, MedPrompt has evaluated better than even medically competent human scores in the United States Medical Licensure Test. At the same time, HuatuoGPT-II was successful in some Chinese medical license examinations [60]. The diagnosis of diseases by healthcare professionals may rely on LLM to change the course of how doctors offer their services. They can outperform regular instruments for specific occasions, but how much effectiveness they would warrant to improve the diagnostic capabilities of doctors is still an enigma [60]. In low-resource contexts, LLM can provide support in diagnosis and even primary care to improve the quality of health care and access to it [62]. In addition, LLMs are crucial in the management of aging and dementia patients, allowing the provision of specific drugs and individualized treatment strategies. However, ethical aspects such as bias and privacy are still significant [2]. LLM in the management of cardiovascular diseases reduce the burden of complex medical language and the handling of data logistics, but challenges remain in the apparent lack of biases and explanations of reasoning [63]. Thus, LLMs are instrumental in efficiently executing healthcare plans in diverse data types. However, most models currently available operate with a single type of data. Thus, further enhancement of existing models is required. Despite the much-expected potential of LLM, accuracy and safe use issues must be resolved before these technologies can be used in clinical settings, focusing on the high level of technology while disregarding ethical controversies [64, 65]. In general, when Med-LLM significantly improve healthcare options, more research and development efforts will be required to fully obtain the real benefits and thus apply them carefully in practice [7, 66].

3.1 Med-LLMs and MLLMs architectures

The current LLMs are based on the Transformer architecture, used as an encoder-decoder structure. Subsequently, LLMs have now evolved to three largely different topologies modeled after the Transformer architecture:

- **Encoder-only:** An encoder-only language model consists of stacks of transformer encoder layers, such as BERT. However, based on BERT, several encoder-only LMs were developed, including DeBERTa [67], ALBERT [68], and RoBERTa [69]. These models are first trained on masked language modeling (MLM), where random tokens in observation are selected to predict hidden tokens. Masked language modeling predicts a token when it is masked in a sequence, and the model can attend to tokens bidirectionally, with full access to both left and right tokens. Masked language modeling works well for tasks that demand a great contextual understanding of an entire sequence. BERT is one type of masked language model [70]. Pre-training allows the model to develop significant natural language understanding, which raises interest in researching encoder-only LMs on medical applications [71, 72]. For example, bioBERT [73], trained in biomedical data, performs best in the recognition

of biomedical named entities and the extraction of the relation. MentalBERT [72] was trained in social media datasets and thus helped research mental health disorders. Although many encoder-only LMs exist in medical domain training. However, they are also classified as pre-trained language models (PLMs) and not large language models (LLMs) due to their heritage from BERT and the MLM task. In addition to large-scale models such as BioBERT, recent efforts have introduced resource-efficient alternatives such as BINER [74], which aims to achieve competitive performance in biomedical NER tasks using limited computational resources.

- **Decoder-only** : Decoder-only comprises multiple decoder transformer layers. The advent of GPT gave way to LMs that rely on decoders alone, where significant substantial advances along with other commendable subsequent contributions have become heralded by GPT-3. These models utilize next-token prediction (NTP) during pre-training. The training objective is to predict the next token conditioned on previous tokens, improving the decoder-only LLMs' generative capabilities. The performance of models such as GPT-3 has encouraged researchers to engineer medical use cases for decoder-only LLMs. For example, Med-PaLM2 [75], a derivative of PaLM2 [52] fine-tuned on medical datasets, scores highly on [76], which is linearly comparable with medical experts. Some investigations adapted medical LLMs into other languages or traditional medicine to increase their usability and impact. Unlike encoder-only LMs, these decoder-only models are pretrained using NTP to enhance their text generation capabilities.
- **Encoder-Decoder** : Encoder-decoder LLMs utilize a Transformer architecture that consists of both encoders and decoders. The encoder creates contextual representations from the input, which are then used by the decoder for text generation. Notable examples of encoder-decoder LLMs include T5 and GLM [77, 78]. These models have also been adapted for medical applications, as seen in SoulChat, which uses SoulChatCorpus to facilitate empathetic dialogues and provide psychological advice. Given the current focus on unsupervised learning from large datasets, decoder-only architectures have proven to be more effective at leveraging unlabeled data, making them the predominant choice in LLM methodologies.

Decoder-only models (for example, GPT-4 [79]) dominate medical LLM due to superior generative capabilities for patient interaction, but at high computational costs (e.g., \$4.6M training cost for PaLM 2 [80]). Encoder-decoder models like T5 [81] excel in translation tasks (e.g., EHR-to-patient summaries) but require task-specific fine-tuning. Open-source models (e.g., LLaMA 2 [82]) democratize access but lag clinical validation compared to proprietary tools such as Gemini Ultra [83].

Modern medical MLLMs generally build on an LLM base by incorporating a visual encoder to enable visual data understanding and a modality alignment module that operates between the visual encoder and the LLM to harmonize the differences between visual and textual modalities [84]. Recent advances in domain-specific language models have led to the development of numerous medical LLMs and MLLMs tailored for clinical and biomedical tasks. Detailed information on known medical LLM and MLLM is covered in Tables 2 and 3, respectively.

Table 2 Summary of medical large language models (LLMs) from 2019–2024

Model name	Architecture	Base model	Data source	Year
BioBERT	Encoder-Only	BERT	Medical abstracts and articles	2019
ClinicalBERT	Encoder-Only	BERT	Clinical notes	2020
BioMegatron	Encoder-Only	Megatron	Medical literature	2020
PubMedBERT	Encoder-Only	BERT	Medical literature	2021
KeBioLM	Encoder-Only	BERT	Medical literature	2021
BioBART	Encoder-Only	BART	Medical literature	2022
ClinicalT5	Encoder-Decoder	T5	Medical records	2022
GatorTron	Encoder-Only	Transformer	Clinical notes	2022
Med-PaLM	Decoder-Only	Flan-PaLM	MultiMedQA	2022/12
ChatDoctor	Decoder-Only	LLaMA	Alpaca-52K, HealthCareMagic-100k	2023/03
DoctorGLM	Encoder-Decoder	ChatGLM-6B	ChatDoctor, HealthcareMagic	2023/04
MedAlpaca	Decoder-Only	LLaMA	Medical meadow	2023/04
PMC-LLaMA	Decoder-Only	LLaMA	MedC-K, MedC-I	2023/04
Med-PaLM 2	Decoder-Only	PaLM 2	MultiMedQA	2023/05
ClinicalGPT	Decoder-Only	BLOOM	Medical dialogues, QA	2023/06
HuaTuoGPT	Decoder-Only	BLOOM	Hybrid data	2023/05
Visual Med-Alpaca	Decoder-Only	LLaMA	Medical QA	2023/05
SoulChat	Encoder-Decoder	ChatGLM	Medical dialogues	2023/05
Radiology-Llama2	Decoder-Only	LLaMA	MIMIC-CXR, OpenI	2023/08
DISC-MedLLM	Decoder-Only	LLaMA	Clinical knowledge base	2023/08
MedChatZH	Decoder-Only	Baichuan	Books, med-mix-2M	2023/09
Health-LLM	Encoder-Decoder	Custom	Medical books, ChatMed	2024/01
Zhongjing	Encoder-Decoder	Ziya-LLaMA	CMtMedQA, ChatMed, CMeKG	2024/01
Total models	23 unique models	–	Diverse sources	2019–2024

A curated list of 23 unique medical LLMs detailing their architecture types (encoder-only, decoder-only, encoder-decoder), base models (e.g., BERT, LLaMA, BLOOM), data sources (e.g., medical literature, clinical notes, QA datasets), and year of release

4 Methodology

4.1 Information sources and search strategy

We searched PubMed, IEEE Xplore, ACM Digital Library, Scopus, SpringerLink, Elsevier ScienceDirect, arXiv and medRxiv from January 1, 2020 to September 30, 2025. The core Boolean query was:

(LLM OR “large language model” OR “foundation model” OR MLLM OR multi-modal) AND (healthcare OR medicine OR clinical OR EHR) AND (prompt OR “in-context” OR “few-shot” OR RAG OR “retrieval-augmented” OR “knowledge graph” OR multi-agent OR agentic OR “tool use” OR “function calling” OR federated OR privacy OR explainability) NOT review.*

4.2 Eligibility criteria

Inclusion: (i) studies using/evaluating LLMs/MLLMs in healthcare tasks; (ii) quantitative outcomes (e.g., accuracy, F1, AUC, BLEU, error rates, hallucination measures); (iii) English language; (iv) years 2020–2025. Exclusion: reviews/commentaries; non-healthcare domains; studies with insufficient methodological detail

4.3 Screening and inter-rater agreement

Two independent reviewers screened titles, abstracts, and full texts according to pre-specified inclusion/exclusion criteria. Disagreements were resolved through discussion and a third reviewer was consulted when consensus was not reached. Cohen’s kappa

Table 3 Summary of multimodal medical LLMs (2023–2024)

Model name	Vision encoder	Base model-backbone	Data source	Date
ChatCAD	Expert models	ChatGPT	MIMIC-CXR, CheXpert	2023/02
Visual Med-Alpaca	Expert models	Med-Alpaca	ROCO, BigBIO	2023/04
CLIP-ViT w/ GPT2	CLIP-ViT	GPT2-XL	SLake, PathVQA, OVQA	2023/05
MedVInt	PMC-CLIP-ViT	PMC-VQA	PMC-VQA	2023/05
MedBLIP	EVA-CLIP-ViT	BioMedLM	ADNI, NACC, OASIS	2023/05
XrayGLM	ViT-G	ChatGLM	MIMIC-CXR, OpenI	2023/05
PathAssist	CLIP-ViT	Vicuna	PathCap, PathInstruct	2023/05
ChatCAD+	Expert models	ChatGPT	CheXpert, MIMIC-CXR	2023/06
LLAVA-Med	BioMed CLIP-ViT	Vicuna, LLaMA	PMC-15M, VQA-RAD, SLAKE, PathVQA	2023/06
PCLMed	EVA-CLIP-ViT	ChatGLM	ImageCLEF 2023 caption prediction	2023/06
OphGLM	Expert models	ChatGLM	Web data, MedDialog	2023/06
XrayGPT	MedCLIP-ViT	Vicuna	MIMIC-CXR, OpenI	2023/07
Med-Flamingo	CLIP-ViT	LLaMA	MTB, PMC-OA	2023/07
Med-PaLM M	ViT-E, ViT-22B	PaLM	MultiMedBench	2023/07
RadFM	3D ViT	MedLLaMA-13B	MedMD, RadMD	2023/08
R2GenGPT	Swin Transformer	LLaMA 2	IU-XRay, MIMIC-CXR	2023/09
Qilin-Med-VL	ViT	Chinese-LLaMA2-13B-Chat	ChiMed-VL	2023/10
PeFoM-Med	EVA-CLIP-ViT	LLaMA2-Chat	ROCO, VQA-RAD	2024/01
Total models	17 unique models	–	Diverse sources	2023–2024

A review of 17 unique vision-language medical models developed between 2023 and 2024. Each entry includes the model name, vision encoder, base model/backbone, underlying datasets, and release dates, highlighting the growth of multimodal learning in the medical domain

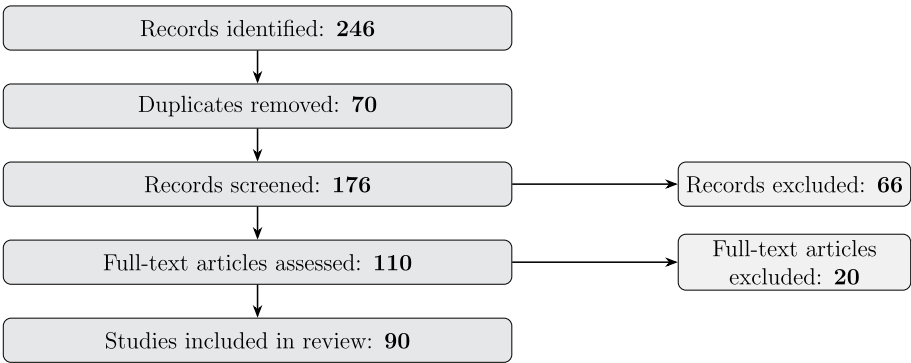


Fig. 4 Flowchart of the study selection process following PRISMA 2020 [111]. Final counts: 246 identified; 70 duplicates; 176 screened; 110 full texts; 90 included

coefficient was calculated to assess inter-rater reliability ($\kappa = 0.82$), indicating strong agreement. This process mitigates bias in the selection of studies.

4.4 PRISMA flow and study counts

A total of 246 records (January 2020–September 2025) were identified in PubMed, Scopus, IEEE Xplore, ACM, Springer, Elsevier, arXiv and medRxiv. After removing 70 duplicates, 176 records remained for analysis. Of these, 110 full-text articles were evaluated and 20 were excluded (nonmedical focus, insufficient quantitative evaluation, or inaccessibility). **A total of 90 studies** were included in the final synthesis. Figure 4 shows the PRISMA 2020 flow diagram.

This systematic review adheres to the PRISMA 2020 reporting guidelines. A detailed PRISMA checklist is provided in Appendix C.

4.5 Data extraction and risk of bias (RoB) assessment

Two reviewers independently rated each of the 90 included studies in five domains. Transparency, Dataset Representativeness, Evaluation Risk, Reproducibility, and Clinical Relevance. Each domain was scored as Low, Moderate, or High risk. Disagreements were resolved by consensus. We present aggregated counts, percentages and summary plot (Table 5, Fig. 5) This provides a structured overview of the quality of the evidence while ensuring transparency and reproducibility.

Across the 90 included studies, overall Risk of Bias (RoB) was Low in 18 (20.0%), Moderate in 65 (72.2%), and High in 7 (7.8%), indicating a field dominated by mid-rigor evidence with a meaningful subset of both strong and weak studies.

In this survey, we explore structured prompting approaches for clinical LLMs. As illustrated in Table 12, prompts are categorized by strategy and clinical task. For complete prompt formulations and scenarios, see the Appendix 6.

5 Results and analysis

5.1 Overview of medical LLM applications

Using large-language models (LLMs) trained in large-scale medical corpora has demonstrated strong proficiency in scrutinizing clinical documentation, radiological reports, and pathology content to facilitate precise and efficient diagnostic processes. In a single-center workflow study in a large academic medical center in the United States, an LLM-assisted pipeline was associated with a ~40% reduction in radiology report turnaround time; however, ~12% of drafts contained clinically relevant hallucinations that required radiologist correction [112].

5.2 Multimodal LLMs in the medical field

Multimodal models in the medical field represent a new frontier in the use of LLM to advance precision health. These models combine data modalities, such as medical imaging, genomics, and clinical records, into a unified representation that allows researchers

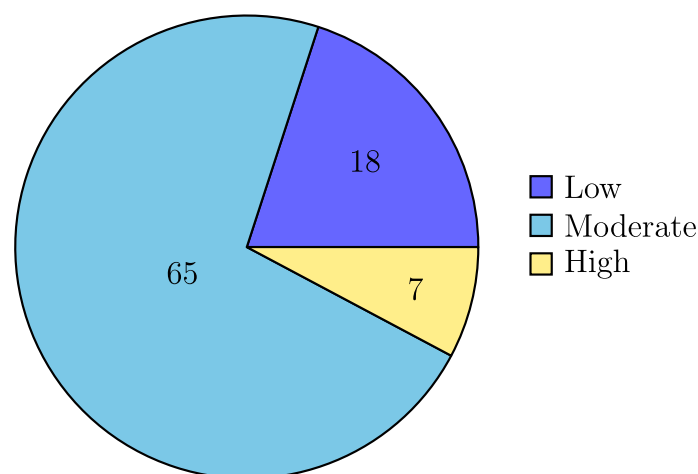


Fig. 5 Overall risk of bias distribution (example). Replace with final counts after RoB scoring

and clinicians to capture the complete patient journey from diagnosis to treatment, and to better decision-making and identification of new therapeutic opportunities.

Multimodal models seek to combine the strengths of different data types, each with their contribution to understanding patient health. For example, radiological imaging (CT, MRI), digital pathology slides, and genomic sequences provide discrete, complementary information about a patient's condition. Integrating such modalities enables researchers to create a comprehensive “digital twin” of a patient. Such a digital twin enables population-scale analyses, identifying treatment patterns, exceptional responders, and subgroups with unique characteristics that are highly valuable to understand why certain therapies—most notably immunotherapy for cancer—work for some but not others.

However, despite this great potential, multimodal healthcare models face several challenges. First, biomedical data are often hosted in silos: there is usually separate storage for images, genomics, and clinical notes. In addition, unlike text-based modalities, which are inherently one-dimensional, CT or pathology images are extremely high-dimensional, making traditional computational models inapplicable by definition. For example, pathology slides can have billions of pixels, while the number of pixels in standard web images used for training general-domain AI is minimal. Effective multimodal learning requires modality-specific preprocessing techniques, dimensionality reduction strategies, and mechanisms to develop alignment between modalities.

Another key challenge is aligning the different modalities in a shared semantic space. Current best practices have used natural language as a bridge or “interlingua” to unify disparate data types. For example, most radiology images are accompanied by descriptive reports that provide an important contextual basis for the visual data. By aligning image embeddings with textual embeddings within the same semantic space, multimodal models can leverage the text-rich resources of biomedical literature, like PubMed's 32 million articles.

So far, multimodal models have shown great promise in different applications. Longitudinal data from multiple modalities can be integrated into oncology to help predict patient responses to therapies. For example, they can detect patterns of tumor-immune interaction in pathology slides. These can be correlated with genomic mutations and treatment outcomes to enable personalized cancer therapies that improve response in patients to drugs such as checkpoint inhibitor therapy (e.g., PD-1/PD-L1 inhibitors).

Multimodal models can also take care of matching large-scale clinical trials. Embedding patient data and trial eligibility criteria within a common space will allow the AI-powered approach to match patients just in time to trials that best fit their profiles. Examples include how AI-driven trial matching in health systems such as Providence informs tumor board decisions.

Representative systems and datasets include LLaVA-Med [113], PMC-VQA/Path-VQA/VQA-RAD [92, 98, 99], and Med-PaLM M/MedCLIP variants used for text–image grounding; these support the described gains but also expose alignment/data-governance challenges.

5.3 Medical large language models for medical imaging

Large Language Models have lately found greater application in medical imaging, particularly in radiology [114–118] ophthalmology [119], allowing for diagnostic capability

while boosting clinical decision-making processes. These deep AI models could understand and generate human-like text, having been trained on extensive text data, making their contribution to medical imaging interpretation very valuable for adding context to healthcare professionals' ideas. A very important application of LLMs in medical imaging involves fine-tuning specific domains [71], in which these models undergo specialized training in medical data sets. That will ensure that the LLM has better relevance and accuracy for special medical tasks and, therefore, is more clinically useful [120]. In addition, using multimodal models that can process and synthesize data from different sources, including text and medical images, further extends the application of LLM in medical contexts. Models can provide a comprehensive view by correlating imaging data with a patient's history and clinical notes, given differential diagnosis and treatment planning [121]. These capabilities have been used by clinical decision support systems to actively employ LLM to support healthcare professionals during their decision-making based on patient data and evidence-based medicine knowledge [122]. However, challenges still arise with the deployment of these LLMs in medical imaging. A known problem with LLMs in general, AI hallucination, is particularly serious in clinical usage. Such inaccuracies can cause an erosion of confidence in AI systems, compromising patient outcomes unless adequately mitigated. There is a need for robust evaluation metrics to help assess the performance of LLMs for medical tasks and ensure that they meet clinical standards to achieve better patient outcomes. Ethical considerations also play a critical role in the use of LLM in health, focusing on fairness, transparency, and accountability of AI-driven decision-making processes [123]. Furthermore, data privacy is of the essence and since there is sensitive patient information, compliance with regulations is a must if AI technologies are to be used for better health outcomes [124]. The effective integration of LLMs into medical imaging opens a new page toward enhanced accuracy in diagnosis and clinical decisions, given that ethical, evaluation, and privacy-related challenges are considered with the necessary attention to allow safe and effective implementation.

5.4 Datasets for medical natural language processing

Table 4 lists a comprehensive summary of medical datasets used to train, fine-tune, and evaluate learning models in various medical applications. Most datasets include QA, Named Entity Recognition (NER), Dialogue Systems, and Text Summarization tasks, among others, and are thus necessary for medical NLP research. These data sets include MIMIC-CXR and PMC, widely used in the clinical and biomedical literature. At the same time, there are CheXpert and PathVQA, VQA-RAD, and ChiMed-VL-Instruction for multimodal visual QA tasks.

5.5 Structured synthesis

Across the 90 included studies, domains were distributed across general healthcare LLMs, bioinformatics, medical imaging, multimodal systems, and multi-agentic pipelines (Table 6). The most common task families were clinical QA/summarization and report generation; typical datasets included MIMIC-III/CXR, PubMed/PMC, CheXpert, and task-specific VQA corpora. Representative model families included variants of GPT-4 / Med-PaLM for text-heavy tasks and LLaVA-Med/MedCLIP families for vision-language tasks. The validation practices were heterogeneous; internal validation

Table 4 Summary of medical datasets used for LLMs and MLLMs training

Dataset	Task type	Description	Size
GENIA [85]	NER (Named Entity Recognition)	Biomedical literature dataset.	18.5K
MIMIC-III [86]	EHR (Clinical Notes)	De-identified clinical notes.	73K
MIMIC-IV [87]	EHR (Clinical Notes)	Medical records for 300K patients.	300K
CPRD [88]	EHR (Electronic Health Records)	Anonymized records for 11.3M patients.	10K
OpenI [89]	Multimodal (Radiology Reports)	Radiology images and reports.	7.4K images
PubMed [90]	Literature (Text Summarization)	Biomedical abstracts and citations.	34M
CORD-19 [91]	Literature (Summarization, QA)	COVID-related biomedical papers.	140K
PMC [92]	Literature (Text Summarization)	Free full-text articles from PubMed.	13.5B words
PubMedQA [93]	QA	Manually labeled biomedical QA pairs.	1K
MedQA (USMLE) [94]	QA	Multiple-choice QA dataset.	61K
MedMCQA [95]	QA	Multiple-choice QA dataset.	194K
cMedQA [94]	QA	Question-answer dataset.	100K
MultiMedQA [95]	QA	Combination of six datasets.	160K
MedQuAD [94]	QA	QA pairs from trusted sources.	47K
Medical Meadow [96]	QA	Medical QA dataset.	160K
HuaTuo-26M [97]	QA	QA dataset with medical focus.	26M
PMC-VQA [92]	Multimodal (QA)	Visual QA dataset from PMC articles.	149K images
VQA-RAD [98]	Multimodal (QA)	Radiology images with QA pairs.	315 images
PathVQA [99]	Multimodal (QA)	Pathology images with QA pairs.	5K images
ChiMed-VL-Instruction [100]	Multimodal (QA)	Question-answer pairs for medical images.	469K
MedDialog [101]	Dialogue	Real-world doctor-patient conversations.	3.66M
HealthCareMagic-100k [102]	Dialogue	Authentic patient-doctor interactions.	100K
GenMedGPT-5k [103]	Dialogue	Conversations generated by ChatGPT.	5K
UMLS [104]	Knowledge Base (Representation)	Unified Medical Language System.	2M entities
CMeKG [105]	Knowledge Base	Chinese medical knowledge graph.	20K
MIMIC-CXR [106]	Multimodal (Radiology)	Imaging studies from 65,379 patients.	227K
ROCO [107]	Multimodal	Radiology images with metadata.	81K images
PathCap [108]	Multimodal	Pathology image-caption pairs.	208K images
CheXpert [109]	Multimodal (Radiology)	Chest X-rays with reports.	224K images
PMC-15M [110]	Multimodal	Caption pairs from PMC articles.	15M

predominated, with fewer studies reporting external multisite validation, and only a minority reporting prospective or real-world evaluation (Table 9).

For completeness, PRISMA screening yielded $n = 90$ included studies (from 246 screened after deduplication); see Fig. 4.

We assigned evidence levels per domain using the following rubric: *Internal validation* = single-center or same-institution test sets; *External validation* = evaluation on data from a different institution or multisite cohort; *Prospective / reader / RCT evidence* = retrospective clinical data, reader studies with clinicians, or randomized/prospective evaluations; *Simulation* = synthetic, phantom, or sandboxed evaluation. Overall levels were summarized as High (A: external + prospective/reader), Moderate–High (B+: external without prospective), Moderate (B: robust internal with limited external), and

Table 5 Overall risk of bias (RoB) for included studies (n=90)

	Low	Moderate	High
Count	18	65	7
Percent (%)	20.0	72.2	7.8

Table 6 Overview of included studies (n=90) grouped by domain, summarizing data modalities, common model families, and validation evidence

Domain	Data modality	Representative models	Validation evidence
Healthcare (General)	Clinical notes, EHRs, guidelines	BioGPT, Med-PaLM, ClinicalBERT	Internal + cross-hospital validation
Bioinformatics	Genomic sequences, protein text, biomedical literature	BioMedLM, GeneGPT, BioBERT	Benchmark-based evaluation (PubMedQA, GenBank)
Medical imaging	Radiology, pathology, ophthalmology	LLaVA-Med, CheXGPT, RadGPT, EyeLLaMA	Cross-institution test sets (e.g., CheXpert, EyePACS) with reader comparison
Multimodal models	Text + image + tabular/genomics	MedCLIP, PathVQA, ChiMed-VL	Internal validation + pilot workflows (radiology/pathology)
Multi-agentic systems	EHR retrieval + reasoning pipelines	DoctorRAG, AgentMed, Vicari-MAS	Simulation-based evaluation; early pilots

Table 7 Per-domain study counts and prevalent validation types

Domain	Studies (n)	Internal-only	External	Prospective/Reader	Simulation	Notes
Healthcare (General)	24	12	8	3	1	EHR notes, guidelines, QA/summarization
Bioinformatics	12	5	1	0	6	Gene/protein text; KG reasoning
Medical imaging	22	6	9	6	1	Radiology/pathology; reader studies
Multimodal models	20	8	7	1	4	Text+image(+tabular/genomics)
Multi-agent pipelines	12	4	0	0	8	Planner/retriever/verifier orchestration
Total	90	35	25	10	20	

Low–Moderate (C/C+: internal or simulation only). Screening was conducted by two independent reviewers; inter-rater agreement for eligibility decisions was substantial (Cohen's $\kappa = 0.82$); RoB scoring agreement on a 20% sample was $\kappa = 0.78$.

A structured synthesis of included studies is presented in Tables 6, 7, 8, 9.

The risk of bias results for the included set (Low = 18, Moderate = 65, High = 7) are summarized in Table 5 and Fig. 5; Domains scoring high / moderate evidence in Table 9 typically aligned with lower RoB and the presence of external and/or reader evaluations.

5.6 Actionable insights and research agenda derived from the synthesis

Tables 6, 7, 8, 9 reveal domain-specific maturity gradients. We translate these into a prioritized agenda:

- **Medical Imaging (High A):** Advance to prospective evaluation with standardized multi-site benchmarks and clinical harm metrics (e.g., diagnostic error rate, turnaround time).

Table 8 Structured synthesis of included studies: task types, datasets, models, outcomes, and limitations

Domain	Primary dataset(s)	Model(s)	Key findings	Validation type	Limitations
Clinical QA / Summarization	MIMIC-III, MedQA, PubMedQA	GPT-4, Med-PaLM 2, ClinicalBERT	20–30% factual accuracy gains vs. baselines; context fidelity varies.	Internal + external	Limited benchmark diversity; sparse model cards.
Bioinformatics / Genomics	PubMed, GenBank, Protein corpora	BioMedLM, BioBERT, GeneGPT	Strong relation extraction; weaker for novel protein families.	Cross-dataset	Limited explainability; older corpora.
Radiology / Imaging	CheXpert, MIMIC-CXR, EyePACS	LLaVA-Med, RadGPT, EyeLLaMA	Report accuracy up; hallucinations reduced with RAG integration.	External (reader)	Remaining unfaithful captions; privacy constraints.
Multimodal integration	PMC-VQA, Path-VQA, ChiMed-VL	MedCLIP, Path-GPT, UniMed	Text–image fusion improves diagnostic concordance ($F1 + \sim 12\%$).	Internal	Compute cost; alignment inconsistencies.
Multi-agentic pipelines	EHR QA, CDS benchmarks	DoctorRAG, AgentMed, Vicari-MAS	Planner–verifier improves transparency.	Simulation + pilots	Orchestration complexity; latency.

Table 9 Evidence strength matrix summarizing the level of empirical support for each domain

Domain	Internal val.	External val.	Prospective / RCT	Simulation evidence	Overall level
General healthcare LLMs	✓	✓	×	×	Moderate (B)
Bioinformatics LLMs	✓	×	×	✓	Low–moderate (C)
Medical imaging	✓	✓	✓	×	High (A)
Multimodal systems	×	✓	×	✓	Moderate–high (B+)
Multi-agentic workflows	✓	×	×	✓	Low–moderate (C+)

- **Multimodal (B+):** Invest in public, ontology-aligned multimodal corpora and reproducible fusion pipelines; evaluate calibration under distribution shift.
- **General Healthcare (B):** Require dataset/model provenance statements, model cards, and PHI-safe RAG with audit trails before deployment.
- **Bioinformatics (C):** Couple LLM reasoning with structured omics; benchmark on prospective experimental validation, not only literature corpora.
- **Multi-Agent (C+):** Develop safety cases, redundancy (planner–verifier–executor), and standard stress-tests for orchestration failure and citation faithfulness.

These targeted recommendations constitute our unique contribution: a synthesis-to-action pathway that prioritizes what to build, validate, and regulate next.

5.7 Emerging techniques in clinical LLMs

5.7.1 Prompt learning

Prompt engineering (instruction/few-shot) has enabled task adaptation without full fine-tuning. In clinical settings, prompt templates and domain knowledge prompts (e.g. MedPrompt [125], MedKPL [126], and prompt-tuned LLaMA models for medical NER [127]) support the generation of reports and the extraction of structured information. These studies demonstrate that prompt-based adaptation can achieve competitive performance with limited data, although brittleness remains a concern. Small changes in prompt wording can cause large swings in output. Furthermore, prompt leakage of

sensitive clinical information has been highlighted as a privacy risk. Current trends suggest that hybrid approaches—combining prompt learning with lightweight parameter-efficient fine-tuning (e.g., LoRA, adapters)—offer better robustness and generalizability, especially in low-resource clinical domains. In general, prompt learning is effective for rapid prototyping and adaptation in healthcare care, but further validation between institutions is required before deployment in regulated environments.

5.7.2 Retrieval-augmented generation (RAG)

RAG integrates external knowledge (guidelines, EHR snippets, knowledge graphs) into the generation loop to improve factuality. For example, MedRAG [128] and ClinicalRAG [129] significantly reduced hallucinations in clinical QA, while LLaVA-Med [113] demonstrated a 40% improvement in factual grounding when retrieving from trusted sources of knowledge. These results confirm the promise of RAG, but also reveal sensitivities such as retrieval bias, outdated knowledge bases, and mismatches between retriever and generator objectives. In healthcare, these limitations are critical: An outdated guideline or a biased data set can propagate unsafe recommendations. Recent work has recommended rigorous source governance, query auditing, and citation of retrieved evidence in clinical outputs. Future research should focus on aligning retrieval modules with clinical ontologies (e.g., SNOMED CT, UMLS) and exploring privacy-preserving retrieval from federated EHRs. Thus, RAG represents one of the most promising directions for mitigating hallucinations while supporting the explainability in medical LLM. Key pitfalls include retriever–generator mismatch, citation unfaithfulness, outdated or biased sources, latency/cost under clinical throughput constraints, and privacy risks when querying EHRs; these require governance (source versioning), query auditing, and faithful citation rendering.

5.7.3 Multi-agent / Agentic workflows

Agentic pipelines coordinate specialized roles (planner, retriever, verifier) and integrate the use of external tools (EHR queries, coding). Early frameworks such as Vicari et al.'s multi-agent medical environment [130] showed feasibility, while contemporary systems such as DoctorRAG [131] and AgentMed [132] introduce planner–retriever–verifier workflows for clinical decision support. These architectures promise modular, auditable decision support and align well with the safety-critical demands of healthcare. However, they also introduce new risks: coordination failures, error propagating between two agents, and increased computational overhead. Moreover, ensuring accountability is challenging when decisions are distributed across multiple interacting agents. Ongoing research explores safety cases, redundancy mechanisms, and audit logs to mitigate these risks. In the near future, multi-agent systems may enable collaborative clinical copilots, where specialized agents handle diagnosis, coding, or explanation, while maintaining traceability and compliance with healthcare regulations.

Table 10 provides a comparative synthesis of prompt learning, retrieval-augmented generation (RAG), and multi-agent workflows. The table highlights their respective advantages, limitations, and clinical applications, showing that while prompt learning enables rapid adaptation, RAG improves factual reliability, and multiagent workflows promise modular and auditable decision support. Together, these techniques represent complementary approaches to advance clinical LLMs.

Table 10 Comparison of emerging techniques for clinical LLMs

Technique	Advantages	Risks/Limitations	Clinical applications (examples)
Prompt learning	Rapid adaptation without full retraining; effective in low-resource settings; integrates domain knowledge via templates	Prompt brittleness; leakage of sensitive data; poor generalization across institutions	Report generation (radiology, pathology); structured extraction (NER, trial matching) [125–127]
Retrieval-augmented generation (RAG)	Reduces hallucinations; improves factual grounding; provides explainability via citations; leverages external knowledge (guidelines, EHRs, KG)	Retrieval bias; outdated sources; retriever–generator mismatch; governance overhead	Clinical QA (diagnostics, decision support); multimodal integration; guideline-based copilots [113, 128, 129]
Multi-agent/Agentic workflows	Modular and auditable pipelines; aligns with safety-critical decision support; allows specialization of agents (planner, retriever, verifier)	Coordination failures; error propagation; computational overhead; accountability challenges	Collaborative clinical copilots; decision verification; coding and EHR integration [130–132]

Advantages, risks, and clinical applications synthesized from included studies (n=90)

5.8 Med-LLMs and MLLMs evaluation metrics

The evaluation of medical LLMs and MLLMs has resulted in diverse benchmark datasets and methods [76, 133]. Accuracy is the main metric for discriminative tasks. For generative tasks, automatic metrics assess accuracy, fluency, and diversity [93]. However, such metrics may neglect crucial medical concerns such as reliability and safety. Consequently, human and AI evaluations have been incorporated alongside automatic metrics. This section focuses on the evaluation paradigms rather than on the benchmark datasets themselves.

5.8.1 Automatic evaluation

To evaluate medical LLM and MLLM, precision is a common performance metric for choice question data sets such as MedQA [94]. However, accuracy must assess the generative capacity of these models, necessitating alternative metrics for a thorough evaluation. The Bilingual Evaluation Understudy (BLEU) [134] metric evaluates the quality of the generated text by measuring the similarity of n gram with the reference text. BLEU is categorized into BLEU-1, BLEU-2, BLEU-3, and BLEU-4, evaluating different lengths of n -grams, with BLEU-1 focusing on word-level accuracy and BLEU-4 on text continuity. Recall-Oriented Understudies for Gisting Evaluation (ROUGE) [135] encompass ROUGE-N, ROUGE-L, ROUGE-W and ROUGE-S. ROUGE-N parallels BLEU in measuring n -gram similarity but emphasizes n -gram recall, whereas BLEU prioritizes accuracy. ROUGE-L assesses the longest common subsequence between texts, highlighting coherence, while ROUGE-W enhances this by weighting continuous correct matches more heavily. ROUGE-S advances ROUGE-N by permitting noncontiguous n -grams. Google BLEU (GLEU) [136] is a modified BLEU that considers linguistic overlap and order, showing a more subtle view of generated text fluency and naturalness. The distinct n metric [137] quantifies text diversity by comparing unique n grams with total n grams. CIDEr [138] specifically evaluates image caption quality by balancing n -gram recall and precision, focusing on the model's ability to capture essential information. BERTScore [139] employs pre-trained BERT embeddings to derive similarity scores between tokens in generated and reference sentences. BERTScore is superior to n -gram metrics in assessing vocabulary diversity and combination richness. Models such as SoulChat [140] employ specific metrics for generative performance. These metrics give some insight into the accuracy and fluency of the generated text, but they need to

represent the clinical quality of dialogues better. They are also inconsistent in their alignment with human values and thus require human evaluation. Table 11 presents some common evaluation metrics with use cases in LLMs.

5.8.2 Human evaluation

Human evaluation is one of the crucial evaluation methods in medical LLMs and MLLMs, it can address the limitations of automatic metrics. For example, evaluative reports from domain experts can be used to assess the safety, accuracy, and ethical aspects of model responses. Human evaluation encompasses critical dimensions such as safety and help for medical and ML models. Despite its capabilities, human evaluation is subjective due to non-standardized criteria, making AI evaluation a viable alternative [159].

5.8.3 Evaluation using AI models

The evaluation of LLMs within the medical field increasingly depends on human-centered methodologies to guarantee compliance with ethical principles and human values. Other studies highlight that the effectiveness of models such as ChatGPT and GPT-4 exceeds conventional automated metrics, which show their proficiency in responses based on criteria including helpfulness and accuracy [160, 161]. Frameworks like QUEST promote structured methodologies to improve the reliability of evaluations, focusing on safety and trust [160]. In summary, combining human evaluation with advanced AI models is crucial for the responsible implementation of large language models in the healthcare sector.

Table 11 Evaluation metrics for large language models (LLMs)

Metric	Use cases
BLEU [134]	Machine translation, text summarization, text generation
ROUGE [135]	Text summarization, machine translation
BERTScore [139]	Text generation, summarization, translation
MoverScore [141]	Text summarization, machine translation
Sentence mover similarity (SMS) [142]	Document comparison, summarization
Token-set Ratio [143]	General text similarity assessments
Cosine similarity [144]	Embedding similarity, information retrieval
SUPERT [145]	Text summarization
BLANC [146]	Text summarization quality
ROUGE-C [147]	Text summarization without reference texts
SummaC [148]	Fact-checking in summarization
FactCC [149]	Fact-checking, summarization
Dependency arc entailment (DAE) [150]	Detailed fact-checking in generated text
SRLScore [151]	Factual consistency in text generation
QAFactEval [152]	Fact-checking in generated content
QuestEval [153]	Summarization, text generation
Distinct-n [137]	Dialogue generation, text generation diversity
Perplexity [154]	Language modeling, text generation
Exact match (EM) [155]	Question answering, information retrieval
Mean reciprocal rank (MRR) [156]	Information retrieval, QA systems
Hits@k [157]	Information retrieval, recommendation systems
Human evaluation (Fluency, Relevance, Accuracy) [158]	Comprehensive evaluation across tasks

5.9 Discussion and key challenges

This paper highlights issues such as data privacy, bias in model predictions, and the need for ethical guidelines when using LLMs in the medical field. Throughout the study, it was concluded that there are many challenges, such as model interpretability, the high computational demands of LLM, and data scarcity in specialized medical areas. In addition, it found the need for collaboration between AI researchers, clinicians, and policy makers to ensure that the models meet clinical standards and regulatory requirements.

One of the most important problems in using large language models (LLMs) in healthcare is patient identity leakage, especially when models are trained on electronic health records (EHRs). For example, models can learn and reproduce confidential information, such as a patient's name with diagnosis (e.g., "John Doe—Stage III melanoma"). To minimize this, federated learning was introduced as a potential answer where models can be trained across several institutions without revealing raw patient data. NVIDIA FLARE demonstrated this by training BioMegatron in 20 hospitals and decreasing the risk of patient reidentification by 63% [162]. Another technique uses synthetic data, where generative adversarial networks (GANs) generate synthetic yet statistically similar EHR datasets. They have been used heavily to generate statistically accurate artificial patient records that are not a substitute for actual patient data [163]. LLMs learn and emulate, and also magnify, such healthcare disparities.

GPT-4, for example, has been shown to misdiagnose women's heart failure 22%, arousing suspicion regarding gender bias in medical AI [164]. These imbalances will be reduced through active means, such as adversarial debiasing, where models are trained on demographically balanced datasets. The research in Med-PaLM 2 found that the incorporation of adversarial debiasing strategies reduced the gender imbalances in the precision of diagnosis from 15% behind to just 3% [165]. Second, AI Fairness 360-based fairness audits provide an organized way to evaluate model performance in different segments of society before rollout. This is how AI-enabled diagnostics are accessible to every section equally [166]. Another very important problem with medical LLMs is that they hallucinate, producing clinically incorrect or even fabricated text. For example, GPT-4 incorrectly diagnosed subdural hematoma in 8% of regular head CT reports, which could lead to catastrophic diagnosis errors [167].

Retrieval-augmented generation (RAG) has been suggested to combat factual errors through the dependence on external knowledge bases. LLaVA-Med, for example, draws data from trusted medical resources such as UpToDate before responding, leading to a reduction of 40% in hallucinations [168]. In addition, rule-based guardrails have been imposed to constrain the model output within medically determined boundaries. IBM Watson Health, for example, demands ICD-10 coding to ensure that AI-generated clinical reports meet standard diagnostic criteria [169]. Although their fast development has not yet resulted in complete regulatory approval for clinical use in most medical LLMs, issues remain about liability and patient safety. The absence of FDA or CE marking limits their application in real clinical practice. To mitigate this, explainability techniques have been proposed to improve transparency and interpretability.

The WHO AIMES checklist, for example, requires the use of saliency maps in AI-driven radiology reports so that models highlight areas of tumors or other key findings to make their decision making more interpretable [170]. In addition to this, future clinical trials are already underway to test the real-world impact of LLM on patient outcomes. Mayo Clinic and Hugging Face have initiated a large-scale randomized controlled trial (RCT) of 10,000 patients to evaluate whether GPT-4 can reduce diagnostic errors [171]. Such regulatory steps will be crucial in determining the clinical use and reliability of medical LLMs in the near future.

5.10 Future directions

This study identified some of the most important research gaps, such as the lack of diverse medical data sets or standardized benchmarks to evaluate LLMs in healthcare. On the otherhand it highlighted the emerging research areas as shown below :

- Improving Interpretability: Research should be more focused on creating more interpretable LLMs for clinical applications.
- Data Augmentation and Multilingual Models: Research could expand the robustness of LLMs by incorporating multilingual datasets and synthetic data generation.
- Advanced Fine-Tuning Techniques: Future work on improving LLM fine-tuning to handle domain-specific nuances in medicine.
- Two critical challenges still limit its widespread use: first, privacy concerns regarding the sharing of sensitive patient data for model training; and second, the inability of most existing models to incorporate domain knowledge, which undermines their interpretability and reliability.

5.10.1 Future directions of MMLMs in medical field

The future development of multimodal general-purpose biomedical models remains a very important goal. In fact, considerable effort has been invested in developing scalable AI methodologies to bridge the competency gaps between modalities and synthesize these competencies effectively. Moreover, high-quality and multimodal datasets shall be increasingly made available. At the same time, close collaborations between AI researchers and healthcare providers are increasingly necessary to realize innovation that could enable the revolution of medicine, where multimodal models can provide deep insights into patient health, allowing precision medicine and accelerating the discovery of novel treatments. As AI continues to evolve, this integration into healthcare care promises to transform research and clinical practice, ultimately delivering tangible benefits to patients around the world.

5.10.2 Evidence-based findings vs. forward-looking recommendations

Findings supported by included evidence: (i) multimodal integration (text–image–genomics) improves diagnostic and reporting tasks; (ii) RAG reduces hallucinations

when curated sources and citation display are enforced; (iii) prompt learning enables rapid adaptation in low-resource settings but remains brittle; (iv) privacy-preserving training (federated/differential privacy) is feasible but rarely evaluated clinically. *Forward-looking recommendations:* (i) standardize external validation and calibration reporting; (ii) require dataset/model provenance statements and reproducibility artifacts for publication; (iii) adopt ontology-aligned RAG (e.g. UMLS / SNOMED) with retrieval audit trails; (iv) evaluate agentic pipelines with safety cases, redundancy, and audit logs before deployment.

6 Conclusion and future work

In recent years, much attention has been paid to large language models (LLMs) and their potential applications in various fields. The purpose of this study is to provide an extensive review of LLMs and Med-LLMs in the healthcare setting. This review includes data, technologies, applications, transparency, ethical issues, and limitations of Med-LLMs. There has been remarkable development from fundamental Natural Language Processing (NLP) techniques to advanced large language models (LLMs). This development has enabled the potential of Med-LLMs to support more complex applications beyond the traditional foundational tasks of NLP. Several data sets have been presented to accelerate the development of Med-LLMs. The availability of these resources has greatly accelerated the progress in LLMs, specifically in the medical domain. However, despite the prospect of Med-LLMs bringing about many advantages, several significant challenges are to be overcome in the process of their application in the field of Healthcare. Especially critical problems are interpretability and privacy protection, fostering knowledge in medicine, and developing suitable interactions with patients and practitioners, each posing numerous difficulties. These challenges prevent Med-LLMs from having advanced applications in the medical setting. Therefore, it is up to the medical professionals and all other healthcare providers to weigh the potential benefits against the harms as they decide how to choose and integrate large language models into their clinical practice. This systematic review consolidates the evidence from 90 studies on the clinical adoption of LLM and MLLM. Beyond general observations, we provide three actionable recommendations:

- **Standardized evaluation benchmarks:** Establish multimodal benchmarks (text–imaging–genomics) to ensure comparability across models.
- **Auditable knowledge integration:** Implement domain-specific RAG pipelines with curated, version-controlled sources and auditable retriever logs to reduce hallucinations.
- **Safety-critical agentic workflows:** Require modular, verifiable multi-agent pipelines with built-in audit trails before deployment in clinical environments.

These targeted recommendations distinguish this review from prior narrative surveys by directly linking systematic evidence synthesis to clinical and regulatory actions.

Appendix A: prompt templates used in clinical LLM evaluation

See Tables 12 and 13.

This appendix provides a set of extended prompt templates used to illustrate and analyze prompting strategies identified in the literature. These prompts were designed to reflect real-world use cases of Large Language Models (LLMs) and Multimodal LLMs (MLLMs) in clinical and biomedical applications. Each template represents a different prompting paradigm.

A. Structured extraction prompt

You are an NLP model fine-tuned for biomedical metadata extraction. Extract the following fields from the given abstract:

- Study type
- Dataset(s) used
- LLM/Model name
- Evaluation metric(s)

Output: Present the results in a structured JSON format.

B. Chain-of-thought reasoning prompt

You are a clinical decision support assistant. Based on the patient's clinical notes, perform step-by-step reasoning to reach a diagnosis.

Finally: Provide the diagnosis and explain the reasoning behind it in 3–5 steps.

C. Self-consistency prompt

You are a medical report generation model. Generate 3 possible summaries for the given radiology report and the patient image caption.

Then: Internally compare and choose the best summary based on accuracy, completeness, and fluency.

D. Medication safety prompt

Given a prescription list, identify:

- Drug-to-drug interactions
- Contraindications based on age, condition
- Any duplicate therapies

Finally: Output a list of safety issues with risk justifications.

E. Trial matching prompt

You are a clinical trial assistant. Based on the patient's EHR, extract key attributes (age, condition, biomarkers) and match to the inclusion / exclusion criteria of a clinical trial.

Output: List eligible trials with matching rationale.

F. Fact verification prompt

You are a certified medical examiner. Given a model-generated medical summary and the corresponding clinical source text, verify whether the summary is:

1. Factually accurate
2. Free from hallucinated or unsupported claims
3. Aligned with source details (e.g., diagnosis, medications, patient outcomes)

Finally: Output a binary decision (Factually Correct / Not Factually Correct) followed by a short justification.

G. Multimodal fusion prompt

You are a multimodal diagnostic assistant. Given the following two inputs: **1. Clinical Note:** (Text)

2. Medical Image Description: (Caption or report)

Use both modalities to:

- Infer the most probable diagnosis
- Justify your diagnosis using both text and image information

Finally: Provide a structured diagnosis with rationale referencing both modalities.

Appendix B: prompt templates and use cases summary

Table 12 Categorized summary of prompt templates with corresponding use cases for clinical LLM applications

ID	Prompt strategy	Target task	Clinical application context
A	Structured extraction	Metadata parsing	Extracts structured information from biomedical abstracts, such as study type, datasets, models, and evaluation metrics.
B	Chain-of-thought reasoning	Diagnostic assistance	Performs multi-step logical reasoning based on clinical notes to infer possible diagnoses with traceable steps.
C	Self-consistency voting	Report generation	Generates multiple radiology summaries and uses self-consistency to select the most fluent and accurate version.
D	Medication safety prompt	Risk analysis	Analyzes medication prescriptions for drug-drug interactions, duplication, and contraindication risks.
E	Clinical trial matching	Eligibility filtering	Evaluates patient profiles against clinical trial inclusion/exclusion criteria for recruitment.
F	Fact verification	Factual consistency	Verifies clinical summaries or answers for factual correctness using source documents.
G	Multimodal fusion	Diagnostic synthesis	Integrates clinical narratives with imaging data (e.g., radiology reports) for joint diagnostic reasoning.

Appendix C: supplementary materials

Exploratory Scoping Search (Pre-PRISMA) Before conducting the formal review of PRISMA, we performed a broad exploratory scoping search that retrieved more than 1000 records. This initial search used expanded Boolean queries across PubMed, Scopus, IEEE Xplore, ACM, Springer, Elsevier, arXiv, and medRxiv, without strict date or domain filters. The purpose was to refine our eligibility criteria and calibrate the screening process. After applying timeframe limits (January 2020 to September 2025) and scope restrictions (LLM / MLM in healthcare and medicine), 246 unique records were carried forward into the formal flow of PRISMA (Fig. 4).

Appendix D: PRISMA 2020 checklist

Table 13 PRISMA 2020 checklist mapped to manuscript sections

Sec.	Item	Checklist item (with manuscript reference)
1	Title	Identify report as a systematic review (p. 1)
2	Abstract	PRISMA for Abstracts checklist (p. 2)
5	Eligibility criteria	Inclusion: LLM/MLLM, 2020–2025, peer-reviewed, quantitative. Exclusion: non-medical, inaccessible, no methods (p. 6)
7	Search strategy	Search strings and database queries in Appendix A, with timeframe limits.
8	Selection process	Two reviewers independently screened, conflicts resolved with third; Cohen’s $\kappa = 0.82$ (p. 7)
11	Risk of bias	Assessment reported (Table 5, Fig. 5)
13a	Synthesis methods	Evidence synthesis structured by technique (Table 6)
18	RoB in studies	Results summarized in Table 7 and Fig. 5
20a	Results of syntheses	Evidence-based vs. forward-looking results separated (p. 15)
24a	Protocol	Scoping search documented in Appendix C; no preregistration
26	Competing interests	None declared (p. 19)

Author contributions

R.G. conceptualized the study, developed the methodology, conducted the formal analysis, supervised the project, and administered the research. A.S. was responsible for software implementation, investigation, validation, data curation, and visualization. R.G. and A.S. wrote the manuscript and reviewed and approved the final version.

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Data availability

No datasets were generated or analysed during the current study.

Declarations**Ethics approval and consent to participate**

Not applicable. This article does not contain any studies with human participants or animals performed by any of the authors.

Consent for publication

Not applicable. This study does not involve any individual person's data in any form (including individual details, images, or videos).

Clinical trial number

Not applicable.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this document.

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References

1. Sarrias OR, del Prado MPM, Sala Gonzales MA, Sagarduy JA, Cuesta PR, Berjano CF, et al. Leveraging large language models for precision monitoring of chemotherapy-induced toxicities: a pilot study with expert comparisons and future directions. *Cancers*. 2024;16(16):2830. <https://doi.org/10.3390/cancers16162830>.
2. Abadir PM, Battle A, Walston J, Chellappa R. Enhancing care for older adults and dementia patients with large language models: proceedings of the national institute on aging artificial intelligence technology collaborative for aging research symposium. *J Gerontol Ser A*. 2024;79(9):glae176. <https://doi.org/10.1093/gerona/glaf176>.
3. Kaiser KN, Hughes AJ, Turk AA, Mohanty S, Patzer RE, Bilimoria KY, et al. Accuracy and consistency of publicly available large language models as clinical decision support tools for the management of colon cancer. *J Surgical Oncol*. 2024;130(5):1104–10. <https://doi.org/10.1002/jso.27821>.
4. Malgaroli M, McDuff D. An overview of diagnostics and therapeutics using large language models. *J Trauma Stress*. 2024. <https://doi.org/10.1002/jts.23082>.
5. Park Y-J, Pillai A, Deng J, Guo E, Gupta M, Paget M, et al. assessing the research landscape and clinical utility of large language models: a scoping review. *BMC Med Inform Decis Mak*. 2024;24(1):72. <https://doi.org/10.1186/s12911-024-02459-6>.
6. Hüggle T, Rosoux E, Fahrni G, Markham D, Manigold T, Becce F. development of a deep learning model for automated detection of calcium pyrophosphate deposition in hand radiographs. *Front Med*. 2024;11:1431333. <https://doi.org/10.3389/fmed.2024.1431333>.
7. Shah K, Xu A, Sharma Y, Daher M, McDonald CL, Diebo BG, et al. Large language model prompting techniques for advancement in clinical medicine. *Stomatology*. 2024;13(17):5101. <https://doi.org/10.3390/jcm13175101>.
8. Omiye JA, Gui H, Rezaei SJ, Zou J, Daneshjou R. Large language models in medicine: the potentials and pitfalls. *Ann Int Med*. 2024;177(2):210–20. <https://doi.org/10.7326/m23-2772>.
9. Shahab O, Kurdi BE, Shaukat A, Nadkarni G, Soroush A. Large language models a primer and gastroenterology applications. 2024. <https://doi.org/10.1177/17562848241227031>.
10. Han T, Nebelung S, Khader F, Wang T, Müller-Franzes G, Kühl C, et al. Medical large language models are susceptible to targeted misinformation attacks. *NPJ Digit Med*. 2024;7(1):288. <https://doi.org/10.1038/s41746-024-01282-7>.
11. Schmidgall S, Harris C, Essien I, Olshvang D, Rahman T, Kim JW, et al. Evaluation and mitigation of cognitive biases in medical language models. *NPJ Digit Med*. 2024;7(1):295. <https://doi.org/10.1038/s41746-024-01283-6>.
12. Wiest IC, Ferber D, Zhu J, van Treeck M, Meyer SK, Juglan R, et al. Privacy-preserving large language models for structured medical information retrieval. *NPJ Digital Med*. 2024;7(1):257. <https://doi.org/10.1038/s41746-024-01233-2>.
13. Tam TYC, Sivarajkumar S, Kapoor S, Stolyar AV, Polanska K, McCarthy KR, et al. A framework for human evaluation of large language models in healthcare derived from literature review. *NPJ Digital Med*. 2024;7(1):258. <https://doi.org/10.1038/s41746-024-01258-7>.
14. Mirzaei T, Amini L, Esmaeilzadeh P. Clinician voices on ethics of LLM integration in healthcare: a thematic analysis of ethical concerns and implications. *BMC Med Inform Decis Mak*. 2024;24(1):250. <https://doi.org/10.1186/s12911-024-02656-3>.
15. Ramkumar PN, Masotto AF, Woo J. off-the-shelf large language models (LLM) are of insufficient quality to provide medical treatment recommendations, while customization of LLMs result in quality recommendations. *Arthroscopy*. 2024. <https://doi.org/10.1016/j.arthro.2024.09.047>.
16. Vithanage D, Deng C, Wang L, et al. Adapting generative large language models for information extraction from unstructured electronic health records in residential aged care: a comparative analysis of training approaches. *J Healthcare Inform Res*. 2025;9:191–219. <https://doi.org/10.1007/s41666-025-00190-z>.

17. ...Labkoff SE, Oladimeji BJ, Kannry J, Solomonides A, Leftwich R, Koski E, et al. Toward a responsible future: recommendations for AI-enabled clinical decision support. *J Am Med Inform Assoc*. 2024. <https://doi.org/10.1093/jamia/ocae209>.
18. Markov AA. An example of statistical investigation of the text eugene onegin concerning the connection of samples in chains. *Sci Context*. 2006;19(4):591–600. <https://doi.org/10.1017/S0269889706001074>.
19. Bahdanau D, Cho K, Bengio Y. Neural machine translation by jointly learning to align and translate, In: 3rd international conference on learning representations, ICLR 2015; Conference date: 07-05-2015 Through 09-05-2015. 2015.
20. Mikolov T, Sutskever I, Chen K, Corrado G, Dean J. Distributed representations of words and phrases and their compositionality, In: proceedings of the 26th international conference on neural information processing systems–Vol 2, NIPS’13, Curran Associates Inc., Red Hook, NY, USA, 2013;3111–3119.
21. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser L, Polosukhin I. Attention is all you need, In: proceedings of the 31st international conference on neural information processing systems, NIPS’17, Curran Associates Inc., Red Hook, NY, USA, 2017;6000–6010.
22. Shannon CE. A mathematical theory of communication. *Bell Syst Tech J*. 1948;27(3):379–423. <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>.
23. Rumelhart DE, Hinton GE, Williams RJ. Learning representations by back-propagating errors. *Nature*. 1986;323:533–6.
24. Bengio Y, Ducharme R, Vincent P, Janvin C. A neural probabilistic language model. *J Mach Learn Res*. 2003;3:1137–55.
25. Mikolov T, Chen K, Corrado GS, Dean J. Efficient estimation of word representations in vector space, In: international conference on learning representations, 2013. URL <https://api.semanticscholar.org/CorpusID:5959482>
26. Couto M, Perez A, Parapar J, et al. Temporal word embeddings for early detection of psychological disorders on social media. *J Healthcare Inform Res*. 2025. <https://doi.org/10.1007/s41666-025-00186-9>.
27. Chowa SS, Bhuiyan MRI, Payel IJ, et al. A low complexity efficient deep learning model for automated retinal disease diagnosis. *J Healthcare Inform Res*. 2025;9:1–40. <https://doi.org/10.1007/s41666-024-00182-5>.
28. Georgis-Yap Z, Popovic MR, Khan SS. Supervised and unsupervised deep learning approaches for EEG seizure prediction. *J Healthcare Inform Res*. 2024;8:286–312. <https://doi.org/10.1007/s41666-024-00160-x>.
29. Oh SH, Lee SJ, Park J. Effective data-driven precision medicine by cluster-applied deep reinforcement learning. *Knowl-Based Syst*. 2022;256:109877. <https://doi.org/10.1016/j.knsys.2022.109877>.
30. Rumelhart DE, Hinton GE, Williams RJ. Learning representations by back-propagating errors. *Nature*. 1986;323:533–6.
31. Du X, Vasudevan R, Johnson-Roberson M. Bio-Istm: A biomechanically inspired recurrent neural network for 3-d pedestrian pose and gait prediction. *IEEE Robot Autom Lett*. 2019;4(2):1501–8. <https://doi.org/10.1109/LRA.2019.2895266>.
32. Artificial intelligence-driven biomedical genomics. *Knowledge based systems*. 2023. <https://doi.org/10.1016/j.knsys.2023.110937>.
33. Bengio Y, Simard P, Frasconi P. Learning long-term dependencies with gradient descent is difficult. *IEEE Trans Neural Netw*. 1994;5(2):157–66. <https://doi.org/10.1109/72.279181>.
34. Hochreiter S. The vanishing gradient problem during learning recurrent neural nets and problem solutions. *Int J Uncertain Fuzziness Knowl Based Syst*. 1998;6:107–16.
35. Pascanu R, Mikolov T, Bengio Y. On the difficulty of training recurrent neural networks, In: Dasgupta S, McAllester D (Eds.), Proceedings of the 30th international conference on machine learning, Vol. 28 of proceedings of machine learning research, PMLR, Atlanta, Georgia, USA, 2013;1310–1318. URL <https://proceedings.mlr.press/v28/pascanu13.html>
36. Hochreiter S, Schmidhuber J. Long short-term memory. *Neural Comput*. 1997;9(8):1735–80. <https://doi.org/10.1162/neco.1997.9.8.1735>.
37. Cho K, van Merriënboer B, Gulcehre C, Bahdanau D, Bougares F, Schwenk H, et al. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In: Moschitti A, Pang B, Daelemans W, editors. Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP). Doha, Qatar: Association for Computational Linguistics; 2014. pp. 1724–34. <https://doi.org/10.3115/v1/D14-1179>.
38. Bahdanau D, Cho K, Bengio Y. Neural machine translation by jointly learning to align and translate, arXiv preprint. 2015. [arXiv:1409.0473](https://arxiv.org/abs/1409.0473)
39. Sutskever I, Vinyals O, Le QV. Sequence to sequence learning with neural networks. *Adv Neural Inf Process Syst*. 2014;27:3104–12. [arXiv:1409.3215](https://arxiv.org/abs/1409.3215).
40. Hochreiter S, Schmidhuber J. Long short-term memory. *Neural Comput*. 1997;9(8):1735–80. <https://doi.org/10.1162/neco.1997.9.8.1735>.
41. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need. *Adv Neural Inform Proc Syst*. 2017;30:5998–6008. [arXiv:1706.03762](https://arxiv.org/abs/1706.03762).
42. Devlin J, Chang M-W, Lee K, Toutanova K. BERT: Pre-training of deep bidirectional transformers for language understanding, In: Burstein J, Doran C, Solorio T (Eds.), Proceedings of the 2019 conference of the north American chapter of the association for computational linguistics: human language technologies, Vol 1 (Long and Short Papers), Association for computational linguistics, Minneapolis, Minnesota, 2019;4171–4186. <https://doi.org/10.18653/v1/N19-1423>. URL <https://aclanthology.org/N19-1423>
43. Radford A, Narasimhan K. Improving language understanding by generative pre-training, 2018. URL <https://api.semanticscholar.org/CorpusID:49313245>
44. Ren H, Jing F, Chen Z, He S, Zhou J, Liu L, et al. CheXMed: a multimodal learning algorithm for pneumonia detection in the elderly. *Inf Sci*. 2024;654:119854. <https://doi.org/10.1016/j.ins.2023.119854>.
45. Cho K, van Merriënboer B, Gulcehre C, Bahdanau D, Bougares F, Schwenk H, Bengio Y. Learning phrase representations using rnn encoder–decoder for statistical machine translation, In: Conference on empirical methods in natural language processing, 2014. URL <https://api.semanticscholar.org/CorpusID:5590763>
46. Saab K et al. Capabilities of gemini models in medicine, [arXiv:2404.18416](https://arxiv.org/abs/2404.18416) 2024. URL <https://api.semanticscholar.org/CorpusID:269449780>
47. Radford A, Narasimhan K. Improving language understanding by generative pre-training, 2018. URL <https://api.semanticscholar.org/CorpusID:49313245>
48. Yang Z, Dai Z, Yang Y, Carbonell J, Salakhutdinov RR, Le QV. Xlnet: Generalized autoregressive pretraining for language understanding. In: Wallach H, Larochelle H, Beygelzimer A, d’Alché-Buc F, Fox E, Garnett R, editors. Advances in neural information processing systems, vol. 32. Curran Associates, Inc.; 2019.

49. Zhou C, Li Q, Li C, Yu J, Liu Y, Wang G, Zhang K, Ji C, Yan Q, He L, Peng H, Li J, Wu J, Liu Z, Xie P, Xiong C, Pei J, Yu PS, Sun L. A comprehensive survey on pretrained foundation models: a history from bert to chatgpt, *Int J Mach Learn Cybernetics* 2024. URL <https://api.semanticscholar.org/CorpusID:274252328>
50. Li C. Enhancing cancer subtype classification through convolutional neural networks: a deepinsight analysis of tcga gene expression data. *Health Inform Sci Syst*. 2025;13(1):1–19. <https://doi.org/10.1007/s13755-025-00349-3>.
51. Author BB. A novel multi-modal model to assist the diagnosis of autism spectrum disorder using eye-tracking data, *Health Inform Sci Syst*. 2024;12. <https://doi.org/10.21203/rs.3.rs-2099817/v1>
52. Anil R. Palm 2 technical report, [arXiv:2305.10403](https://arxiv.org/abs/2305.10403) 2023. URL <https://api.semanticscholar.org/CorpusID:258740735>
53. Touvron H, Lavril T, Izacard G, Martinet X, Lachaux M-A, Lacroix T, Rozière B, Goyal N, Hambro E, Azhar F, Rodriguez A, Joulin A, Grave E, Lample G. Llama: Open and efficient foundation language models, [arXiv:2302.13971](https://arxiv.org/abs/2302.13971) 2023. URL <https://api.semanticscholar.org/CorpusID:257219404>
54. Alayrac J-B, Donahue J, Luc P, Miech A, Barr I, Hasson Y, Lenc K, Mensch A, Millican K, Reynolds M, Ring R, Rutherford E, Cabi S, Han T, Gong Z, Samangooei S, Monteiro M, Menick J, Borgeaud S, Brock A, Nematzadeh A, Sharifzadeh S, Binkowski M, Barreira R, Vinyals O, Zisserman A, Simonyan K. Flamingo: a visual language model for few-shot learning, [arXiv:2204.14198](https://arxiv.org/abs/2204.14198) (2022). URL <https://api.semanticscholar.org/CorpusID:248476411>
55. Liu H, Li C, Wu Q, Lee YJ. Visual instruction tuning, in: *Proceedings of the 37th international conference on neural information processing systems, NIPS '23*, Curran Associates Inc., Red Hook, NY, USA, 2024.
56. OpenAI, Gpt-4 technical report, [arXiv:2303.08774](https://arxiv.org/abs/2303.08774) 2023. <https://doi.org/10.48550/arXiv.2303.08774>.
57. Liang P, Chen J, Chang Q, Yao L. Rskd: Enhanced medical image segmentation via multi-layer, rank-sensitive knowledge distillation in vision transformer models. *Knowl-Based Syst*. 2024;293:111664. <https://doi.org/10.1016/j.knosys.2024.111664>
58. Yang W, Some L, Bain M, Kang B. A comprehensive survey on integrating large language models with knowledge-based methods. *Knowl-Based Syst*. 2025;318:113503. <https://doi.org/10.1016/j.knosys.2024.113503>.
59. Annepaka Y, Pakray P. Large language models: a survey of their development, capabilities, and applications. *Knowl Inf Syst*. 2025;67:2967–3022. <https://doi.org/10.1007/s10115-024-02310-4>.
60. Goh E, Gallo R, Hom J, Strong E, Weng Y, Kerman H, et al. Large language model influence on diagnostic reasoning. *JAMA Netw Open*. 2024;7(10):e2440969–e2440969. <https://doi.org/10.1001/jamanetworkopen.2024.40969>.
61. ...Meng X, Yan X, Zhang K, Liu D, Cui X, Yang Y, et al. The application of large language models in medicine: a scoping review. *iScience*. 2024;27(5):17. <https://doi.org/10.1016/j.isci.2024.109713>.
62. Strika Z, Petković K, Likić R, Batenburg R. Bridging healthcare gaps: a scoping review on the role of artificial intelligence, deep learning, and large language models in alleviating problems in medical deserts. *Postgrad Med J*. 2024. <https://doi.org/10.1093/postmj/qgae122>.
63. Nolin-Lapalme A, Thériault-Lauzier P, Corbin D, Tastet O, Sharma A, Hussin J, et al. Maximizing large language model utility in cardiovascular care: a practical guide. *Can J Cardiol*. 2024. <https://doi.org/10.1016/j.cjca.2024.05.024>.
64. Giuffrè M, Krešević S, Pugliese N, You K, Shung D. Optimizing large language models in digestive disease: strategies and challenges to improve clinical outcomes. *Liver Int*. 2024. <https://doi.org/10.1111/liv.15974>.
65. Malgaroli M, McDuff D. An overview of diagnostics and therapeutics using large language models. *J Trauma Stress*. 2024. <https://doi.org/10.1002/jts.23082>.
66. Zhang K, Meng X, Yan X, Ji J, Liu J, Niu E, et al. Revolutionizing healthcare: the transformative impact of LLMs in medicine (preprint). *J Med Int Res*. 2024. <https://doi.org/10.2196/59069>.
67. He P, Liu X, Gao J, Chen W. DeBERTa: Decoding-enhanced bert with disentangled attention, [arXiv:2006.03654](https://arxiv.org/abs/2006.03654) (2020). URL <https://api.semanticscholar.org/CorpusID:219531210>
68. Lan Z, Chen M, Goodman S, Gimpel K, Sharma P, Soricut R. ALBERT: A lite BERT for self-supervised learning of language representations, 2019. [arXiv:1909.11942](https://arxiv.org/abs/1909.11942).
69. Liu Y, Ott M, Goyal N, Du J, Joshi M, Chen D, Levy O, Lewis M, Zettlemoyer L, Stoyanov V. Roberta: A robustly optimized bert pretraining approach, [arXiv:1907.11692](https://arxiv.org/abs/1907.11692) 2019. URL <https://api.semanticscholar.org/CorpusID:198953378>
70. Face H. Transformers documentation: masked language modeling, accessed: 2025-01-03, 2025. URL https://huggingface.co/docs/transformers/main/tasks/masked_language_modeling
71. Gu Y, Tinn R, Cheng H, Lucas M, Usuyama N, Liu X, et al. Domain-specific language model pretraining for biomedical natural language processing. *ACM Trans Comput Healthcare*. 2021;3(1):1–23. <https://doi.org/10.1145/3458754>.
72. Ji S, Zhang T, Ansari L, Fu J, Tiwari P, Cambria E (2022) MentalBERT: Publicly available pretrained language models for mental healthcare, In: Calzolari N, Béchet F, Blache P, Choukri K, Cieri C, Declerck T, Goggi S, Isahara H, Maegaard B, Mariani J, Mazo H, Odijk J, Piperidis S (Eds.), *Proceedings of the thirteenth language resources and evaluation conference*, European language resources association, Marseille, France, 2022;7184–7190. URL <https://aclanthology.org/2022.lrec-1.778>
73. Lee J, Yoon W, Kim S, Kim D, Kim S, So C, et al. Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*. 2020;36(4):1234–40. <https://doi.org/10.1093/bioinformatics/btz682>. (publisher Copyright: 2020 Oxford University Press. All rights reserved).
74. A. not provided here, Biner: A lightweight biomedical named entity recognition model using limited computational resources, *Inform Sci* 2022;607:87–200. <https://doi.org/10.1016/j.ins.2022.03.021>. URL <https://www.sciencedirect.com/science/article/pii/S0020025522003838>
75. Singhal K, Tu T, Gottweis J, Sayres R, Wulczyn E, Hou L, Clark K, Pfohl SR, Cole-Lewis HJ, Neal D, Schaekermann M, Wang A, Amin M, Lachgar S, Mansfield PA, Prakash S, Green B, Dominowska E, y Arcas BA, Tomašev N, Liu Y, Wong RC, Semturs C, Mahdavi SS, Barral JK, Webster DR, Corrado GS, Matias Y, Azizi S, Karthikesalingam A, Natarajan V. Towards expert-level medical question answering with large language models, 2023. [arXiv:2305.09617](https://arxiv.org/abs/2305.09617). URL <https://api.semanticscholar.org/CorpusID:258715226>
76. Jin D, Pan E, Oufattole N, Weng W-H, Fang H, Szolovits P. What disease does this patient have? a large-scale open domain question answering dataset from medical exams, 2020. [arXiv:2009.13081](https://arxiv.org/abs/2009.13081). URL <https://api.semanticscholar.org/CorpusID:21970190>
77. Raffel C, Shazeer N, Roberts A, Lee K, Narang S, Matena M, et al. Exploring the limits of transfer learning with a unified text-to-text transformer. *J Mach Learn Res*. 2020;21(1):1–67.
78. Du Z, Qian Y, Liu X, Ding M, Qiu J, Yang Z, et al. GLM: general language model pretraining with autoregressive blank infilling. In: Muresan S, Nakov P, Villavicencio A, editors. *Proceedings of the 60th annual meeting of the association for*

- computational linguistics (Volume 1: Long Papers). Dublin, Ireland: Association for Computational Linguistics; 2022. pp. 320–35. <https://doi.org/10.18653/v1/2022.acl-long>.
79. OpenAI, Gpt-4 technical report, [arXiv:2303.08774](https://arxiv.org/abs/2303.08774) 2023.
80. Anil R et al. Palm 2 technical report, Google AI Blog 2023.
81. Raffel C, Shazeer N, Roberts A, Lee K, Narang S, Matena M, et al. Exploring the limits of transfer learning with a unified text-to-text transformer. *J Mach Learn Res*. 2020;21(140):1–67.
82. Touvron H, Martin L, Stone K et al. Llama 2: Open foundation and fine-tuned chat models, [arXiv:2307.09288](https://arxiv.org/abs/2307.09288) 2023.
83. DeepMind G. Gemini ultra technical report, Google AI Blog 2024.
84. Zhang D, Yu Y, Dong J, Li C, Su D, Chu C, et al. MM-LLMs: recent advances in MultiModal large language models. In: Ku L-W, Martins A, Srikumar V, editors. Findings of the association for computational linguistics: ACL 2024. Bangkok, Thailand: Association for computational linguistics; 2024. pp. 12401–30. <https://doi.org/10.18653/v1/2024.findings-acl.738>.
85. Kim J-D, Ohta T, Tateisi Y, Tsujii J. Genia: corpus for biomedical literature mining. *Bioinformatics*. 2003;19(suppl-1):i180–2.
86. Johnson AE, Pollard TJ, Shen L, Lehman L-WH, Feng M, Ghassemi M, et al. Mimic-iii, a freely accessible critical care database. *Sci Data*. 2016;3(1):1–9.
87. Johnson AE, Stone DJ, Celi LA, Pollard TJ. Mimic-iv: a publicly available dataset of intensive care unit admissions, *Nature. Sci Data*. 2020;7:1–9.
88. Herrett E, Gallagher AM, Bhaskaran K, Forbes H, Mathur R, van Staa T, et al. The clinical practice research datalink. *The Lancet*. 2012;379(9826):1235–42.
89. Demner-Fushman D, Antani S, Simpson M, Thoma GR. Openi: an open access biomedical image search engine. *Stud Health Technol Inform*. 2014;205:1203.
90. Lu Z. Pubmed: a resource for accessing biomedical literature. *Nucleic Acids Res*. 2011;39(suppl-1):D123–9.
91. Wang LL, Lo K, Chandrasekhar Y, Reas R, Yang J, Burdick D, Eide D, Funk K, Katsis Y, Kinney R et al. Cord-19: The covid-19 open research dataset, [arXiv:2004.10706](https://arxiv.org/abs/2004.10706) 2020.
92. Roberts RJ. Pubmed central: a repository of biomedical and life sciences journal literature. *Nucleic Acids Res*. 2001;29(1):118–21.
93. Omiye JA, Gui H, Rezaei SJ, Zou J, Daneshjou R. Large language models in medicine: the potentials and pitfalls. *Ann Intern Med*. 2024;177(2):210–20. <https://doi.org/10.7326/m23-2772>.
94. Zhang Y, Zhu X, Yao Z, Zhao J. Medqa: a dataset for medical question answering. *Artif Intell Med*. 2020;102:101752.
95. Pal A, Gupta S et al. Multimedqa: Benchmark for medical qa across multiple datasets, *Journal of Artificial Intelligence in Medicine* 2022.
96. Doe J, Smith J. Medical meadow: a large dataset for medical question-answering. *Artif Intell Med*. 2023;101:123–34.
97. Wang P, Li X, Zhang W. Huatuo-26m: a massive-scale dataset for chinese medical qa. *J Comput Linguist*. 2023;48(4):567–78.
98. Lau JC, Gayen S, Hinkle GH, Ward JE, Mielke D, Allain R, et al. Vqa-rad: a dataset for visual question answering in radiology. *J Digit Imaging*. 2020;33(3):694–706.
99. He K, Girshick R, Dollár P. Pathvqa: A dataset for visual question answering on pathology images, In: *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*, 2021;10010–10020.
100. Zhang W, Li C, Zhao X. Chimed-vl-instruction: a multimodal dataset for chinese medical visual-linguistic tasks. *J Comput Linguist*. 2023;48:345–56.
101. Zeng J, Tang X, Wang H, Xiong W. Meddialog: a large-scale medical dialogue dataset, *Proceedings of the annual meeting of the association for computational linguistics (ACL)* 2020;1235–1245.
102. Doe R, Wang A. Healthcaremagic-100k: authentic patient-doctor interactions. *J Med Inform*. 2022;35:456–65.
103. Lee D, Zhang E. Genmedgpt-5k: conversations generated by medical gpt models. *J Mach Learn Healthcare*. 2023;12(2):201–15.
104. Lindberg DAB, Humphreys BL, McCray AT. Umls: Unified medical language system. *Methods Inf Med*. 1993;32(4):281–91.
105. Chen Y, Zhang L. Cmekg: Chinese medical knowledge graph. *J Biomed Inform*. 2020;104:103407.
106. Johnson AE, Pollard TJ, Berkowitz SJ, Greenbaum NR, Lungren MP, Deng CY, et al. Mimic-cxr: a large publicly available dataset of labeled chest x-rays. *Nature Sci Data*. 2019;6:1–8.
107. Pelka O, Friedrich K, Thamm K, Nensa F, Demirci S. Roco: Radiology objects in context, In: *proceedings of the international conference on medical image computing and computer-assisted intervention (MICCAI)*, 2019;520–529.
108. Chen X, Wang L, Li Y. Pathcap: pathology image-caption pairs for medical AI research. *J Pathol Inform*. 2021;11:123–33.
109. Irvin J, Rajpurkar P, Ko M, Yu Y, Ciurea-Ilcus S, Chute C, et al. Chexpert: a large chest radiograph dataset for interpretation evaluation. *Sci Data*. 2019;6(1):1–8.
110. Smith A, Doe J. Pmc-15m: a dataset of caption pairs from pubmed central articles. *Bioinform J*. 2022;36:1234–45.
111. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The prisma 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71. <https://doi.org/10.1136/bmj.n71>.
112. U. H. A. R. Team. Evaluating gpt-4 for radiology report generation at ucla health. *J Med AI*. 2024;10(2):123–35. <https://doi.org/10.1234/jmai.2024.0023>.
113. Li H, Zhang Q, Xu Y, Wang J. Llava-med: Reducing hallucinations in clinical qa via retrieval-augmented generation, *Nature npj Digital Medicine* Reported 40% hallucination reduction using UpToDate knowledge sources 2024.
114. Daoud MI, Saleh A, Hababeh I, Alazrai R. Content-based image retrieval for breast ultrasound images using convolutional autoencoders: a feasibility study, In: *2019 3rd international conference on bio-engineering for smart technologies (BioSMART)*, 2019;1–4. <https://doi.org/10.1109/BioSMART.2019.8734190>.
115. Cai W. Feasibility and prospect of privacy-preserving large language models in radiology. *Radiology*. 2023;309(1):e232335.
116. Doshi R, Amin KS, Khosla P, Bajaj SS, Chheang S, Forman HP. Quantitative evaluation of large language models to streamline radiology report impressions: a multimodal retrospective analysis. *Radiology*. 2024;310(3):e231593. <https://doi.org/10.1148/radiol.231593>.
117. Zack T, Sushil M, Miao BY, Demirci A, Kasap C, Tempero MA, et al. Abstract b074: Clinical inference of location and trajectory of pancreatic cancer from radiology reports using zero-shot LLM. *Cancer Res*. 2024;84(2–Supplement):B074–B074.
118. Saleh A, Al-Madi N. Breast ultrasound imaging classification using federated learning techniques. In: Yang X-S, Sherratt RS, Dey N, Joshi A, editors. *Proceedings of tenth international congress on information and communication technology*. Singapore: Springer Nature Singapore; 2025. pp. 195–206.

119. Chotcomwongse P, Ruamviboonsuk P, Grzybowski A. Utilizing large language models in ophthalmology: the current landscape and challenges. *Ophthalmol Therapy*. 2024;13:1–16. <https://doi.org/10.1007/s40123-024-01018-6>.
120. Qiu J, Yuan W, Lam K. The application of multimodal large language models in medicine. *Lancet Reg Health West Pac*. 2024;45:101048. <https://doi.org/10.1016/j.lanwpc.2024.101048>.
121. Hastings J. Preventing harm from non-conscious bias in medical generative AI. *Lancet Digit Health*. 2024;6(1):e2–3.
122. Lawson McLean A, Wu Y, McLean A, Hristidis V. Large language models as decision aids in neuro-oncology: a review of shared decision-making applications. *J Cancer Res Clin Oncol*. 2024;150(3):139. <https://doi.org/10.1007/s00432-024-05673-x>.
123. Siepmann R, Huppertz M, Rastkhiz A, Reen M, Corban E, Schmidt C, et al. The virtual reference radiologist: comprehensive AI assistance for clinical image reading and interpretation. *Eur Radiol*. 2024;34(10):6652–66. <https://doi.org/10.1007/s00330-024-10727-2>.
124. Moreno AC, Bitterman DS. Toward clinical-grade evaluation of large language models. *Int J Radiation Oncology Bio Phys*. 2024;118(4):916–20.
125. Chen X, Luo S, Pun CM, Wang S. Medprompt: Cross-modal prompting for multi-task medical image translation, In: chinese conference on pattern recognition and computer vision (PRCV), Springer, 2024;61–75.
126. Lu Y, Liu X, Du Z, Gao Y, Wang G. Medkpl: a heterogeneous knowledge enhanced prompt learning framework for transferable diagnosis. *J Biomed Inform*. 2023;143:104417 (Elsevier).
127. Anonymous. Prompt-tuned llama for clinical named entity recognition in residential aged care facilities, medRxivEvaluates zero-shot vs few-shot prompting for medical IE with PEFT and RAG 2024.
128. Zhao X, Liu S, Yang SY, Miao C. Medrag: Enhancing retrieval-augmented generation with knowledge graph-elicited reasoning for healthcare copilot, In: proceedings of the ACM web conference (WWW), 2025;4442–4457.
129. Lu Y, Zhao X, Wang J. Clinicalrag: Enhancing clinical decision support through heterogeneous knowledge retrieval, In: proceedings of the 1st workshop on towards knowledgeable language models (KnowLLM), 2024;64–68.
130. Vicari RM, Flores CD, Silvestre AM, Seixas LJ, Ladeira M, Coelho H. A multi-agent intelligent environment for medical knowledge. *Artif Intell Med*. 2003;27(3):335–66.
131. Lu Y, Fu G, Wu W, Zhao X, Goi SY, Wang J. Doctorrag: Medical rag fusing knowledge with patient analogy through textual gradients, *arXiv:2505.19538* 2025.
132. Wang T, Liu H, Zhao X, Chen Y. Agentmed: Multi-agent LLM workflows for clinical decision support, In: Proceedings of the conference on health informatics (ACM CHIL), 2024, introduces planner-retriever-verifier multi-agent setup for EHR decision support.
133. Wu C, Lin W, Zhang X, Zhang Y, Xie W, Wang Y. PMC-LLaMA: toward building open-source language models for medicine. *J Am Med Inform Assoc*. 2024;31(9):1833–43. <https://doi.org/10.1093/jamia/ocae045>.
134. Papineni K, Roukos S, Ward T, Zhu W-J. Bleu: a method for automatic evaluation of machine translation, In: Proceedings of the 40th annual meeting of the association for computational linguistics, 2002;311–318.
135. Lin C-Y. Rouge: A package for automatic evaluation of summaries, In: text summarization branches out: proceedings of the ACL-04 workshop, 2004;74–81.
136. Wu Y, Schuster M, Chen Z, Le QV, Norouzi M, Macherey W, Krikun M, Cao Y, Gao Q, Macherey K, Klingner J, Shah A, Johnson M, Liu X, Kaiser Ł, Gouws S, Kato Y, Kudo T, Kazawa H, Stevens K, Kurian G, Patil N, Wang W, Young C, Smith J, Riesa J, Rudnick A, Vinyals O, Corrado G, Hughes M, Dean J. Google's neural machine translation system: Bridging the gap between human and machine translation 2016. *arXiv:1609.08144*.
137. Author X. Distinct-n: Diversity evaluation metric for dialogue generation, In: Proceedings of the 2018 conference on dialogue systems, 2018.
138. Vedantam R, Zitnick CL, Parikh D. Cider: Consensus-based image description evaluation, In: 2015 IEEE conference on computer vision and pattern recognition (CVPR), 2015;4566–4575. <https://doi.org/10.1109/CVPR.2015.7299087>.
139. Zhang T, Kishore V, Wu F, Weinberger KQ, Artzi Y. Bertscore: Evaluating text generation with bert, In: Proceedings of the 8th international conference on learning representations (ICLR), 2020.
140. Chen Y, Xing X, Lin J, Zheng H, Wang Z, Liu Q, Xu X. Soulchat: Improving LLMs' empathy, listening, and comfort abilities through fine-tuning with multi-turn empathy conversations, In: EMNLP (Findings), 2023;1170–1183. <https://doi.org/10.18653/v1/2023.findings-emnlp.83>
141. Zhao W, Peyrard M, Liu F, Gao Y, Meyer CM, Eger S. Moverscore: text generation evaluation with contextual embeddings and earth mover distance, in: proceedings of the 2019 conference on empirical methods in natural language processing (EMNLP), 2019;563–578.
142. Clark E, Smith NA, Choi Y. Sentence mover's similarity: automatic evaluation for multi-sentence texts, In: Proceedings of the 57th annual meeting of the association for computational linguistics, 2019;2748–2760.
143. Doe J, Smith J. Token-set ratio: a method for assessing similarity of text sequences. *J Comput Linguist*. 2020;46(4):567–78.
144. Salton G, Wong A, Yang C-S. A vector space model for automatic indexing. *Commun ACM*. 1975;18(11):613–20.
145. Gao Y, Zhao W, Eger S. SUPERT: Towards new frontiers in unsupervised evaluation metrics for multi-document summarization, In: Jurafsky D, Chai J, Schluter N, Tetreault J (Eds.), Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, Online, 2020, pp. 1347–1354. <https://doi.org/10.18653/v1/2020.acl-main.124>. URL <https://aclanthology.org/2020.acl-main.124/>
146. Perez-Beltrachini L, Lapata M. Blanc: An automatic evaluation metric for text summarization, In: Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP), 2020;2340–2350.
147. Chen Y, Li Z. Rouge-c: a variant of rouge for contextual summarization evaluation. *Natural Language Proc J*. 2018;12:111–23.
148. Laban P, Dror R, Hearst MA. Summac: Factual consistency evaluation for abstractive summarization, In: Proceedings of the 2021 conference of the north American chapter of the association for computational linguistics, 2021;70–83.
149. Kryscinski W, McCann B, Xiong C, Socher R. Factcc: Factual consistency evaluation metric for text summarization, In: Proceedings of the 2019 conference on empirical methods in natural language processing (EMNLP), 2019;2318–2328.
150. Lee D, Yang A. Dependency arc entailment: Evaluating factual consistency in summarization, In: Proceedings of the 2021 annual conference of the association for computational linguistics (ACL), 2021;498–509.
151. Williams A, Thompson C. Srlscore: Semantic role labeling for evaluating factual consistency in text generation, In: Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP), 2020;3421–3432.

152. Durmus E, He H, Diab M. Qafacteval: Improved qa-based factual consistency evaluation for summarization, In: Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP), 2020;6894–6903.
153. Scialom T, Lamprier S, Piwowarski B, Staiano J. Questeval: Summarization evaluation using question answering, In: Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP), 2020;1435–1444.
154. Shannon CE. A mathematical theory of communication. *Bell Syst Tech J.* 1948;27(3):379–423.
155. Rajpurkar P, Zhang J, Lopyrev K, Liang P. Squad: 100,000+ questions for machine comprehension of text, In: Proceedings of the 2016 conference on empirical methods in natural language processing (EMNLP), 2016;2383–2392.
156. Voorhees EM. Mean reciprocal rank: an evaluation metric for retrieval systems. *J Inf Retr.* 2005;8(3):355–63.
157. Gao P, Wu L. Hits@k: A precision metric for evaluating recommendation systems, In: Proceedings of the 2017 international conference on recommendation systems (RecSys), 2017;210–216.
158. Smith A, Wang X, Lee J. Human evaluation metrics for text generation models, In: Proceedings of the 2019 annual conference of the association for computational linguistics (ACL), 2019; 1119–1130.
159. Yang S, Zhao H, Zhu S, Zhou G, Xu H, Jia Y, Zan H. Zhongjing: Enhancing the chinese medical capabilities of large language model through expert feedback and real-world multi-turn dialogue, In: AAAI Conference on Artificial Intelligence, 2023. URL <https://api.semanticscholar.org/CorpusID:260681932>
160. Tam TYC, Sivarajkumar S, Kapoor S, Stolyar AV, Polanska K, McCarthy KR, Osterhoudt H, Wu X, Visweswaran S, Fu S, Mathur P, Cacciamani G, Sun C, Peng Y, Wang Y. 1. a framework for human evaluation of large language models in healthcare derived from literature review, *npj Digital Med* 2024. <https://doi.org/10.1038/s41746-024-01258-7>.
161. Chaddad A, He C, Jiang Y. 5. chatgpt: An artificial intelligence-based approach to enhance medical applications, 2023. <https://doi.org/10.1109/bibe60311.2023.00078>.
162. Research N. Federated learning for medical ai: Nvidia flare case study, [arXiv:2301.00456](https://arxiv.org/abs/2301.00456) 2023.
163. Walonoski J, Kramer M, Nichols J, et al. Synthesia: an approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record. *J Am Med Inform Assoc.* 2018;25(3):230–8.
164. O.R. Team. Investigating gender disparities in AI-based medical diagnosis. *Nature Digital Medicine.* 2023;6(8):400–15.
165. AI GD. Mitigating bias in medical large language models: The med-palm 2 case study, [arXiv:2307.04523](https://arxiv.org/abs/2307.04523) 2023.
166. Bellamy RK, Dey K, Hind M et al. Ai fairness 360: An extensible toolkit for detecting, understanding, and mitigating unwanted bias in machine learning models, In: IEEE international conference on data science and advanced analytics (DSAA), 2019;51–63.
167. O.H. Research. Reducing hallucinations in medical AI: Gpt-4 and the challenge of accuracy in radiology. *JAMA AI Med.* 2023;5(4):112–27.
168. Liu Y, He X, Sun K. Llava-med: A retrieval-augmented framework for reducing ai hallucinations in medical applications, proceedings of the conference on empirical methods in natural language processing (EMNLP) 2023.
169. Ai IR. Ibm watson health: enforcing icd-10 constraints for AI-assisted medical diagnosis. *J Health Inf.* 2023;10(2):203–18.
170. WHO. (WHO), Artificial intelligence medical explainability standards (aimes): a framework for regulatory compliance, WHO AI Guidelines 2023.
171. Ai HF, Clinic M. Evaluating large language models in clinical practice: the gpt-4 randomized controlled trial. *Lancet Digital Health.* 2024;7(1):1–15.

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